

Advanced Transport Phenomena

Introduction to transport equations*

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September 12, 2026

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*Please note that these notes have been automatically translated from Italian, so the quality of the translation may not be optimal.

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Notice to readers

Most of the content of this document (including the Figures) has been extracted and rearranged from the following books on Computational Fluid Dynamics:

- **J.H. Ferziger, M. Peric, R.L. Street**, *Computational Methods for Fluid Dynamics*, Springer (2020)
- **H. Versteeg, M. Malalasekera**, *Introduction to Computational Fluid Dynamics - The Finite Volume Method*, Pearson (2007)

1 Introduction

Fluid dynamics studies the evolution and dynamics of fluids, both liquid and gaseous. In general, the distinction between gas and liquids is not so important from the point of view of fluid dynamics as the distinction between compressible flow and incompressible flow. From the point of view of thermodynamics, gases and liquids have, as everybody knows, very different characteristics. In particular, liquids are often treated as if they are incompressible, i.e. even significant changes in pressure are not capable of causing significant variations in their density. Very different is the case of gases, which, on the contrary, are considered compressible fluids. The definition that is used in fluid dynamics is completely different so that it is not surprising that we can consider incompressible flow of a compressible fluid. Fluid dynamics defines as incompressible the flows that do not cause significant variations in density in the fluid. There are several criteria to find out if a particular flow is compressible or not. The simplest to use is based on Mach's number, defined as the ratio between fluid velocity and sound velocity under the same conditions:

$$M = \frac{v}{v_{sound}} \quad (1)$$

If this is sufficiently low, the flow can be considered incompressible. The distinction is very important because the approaches used to study a compressible flow are completely

different from those reserved for incompressible flows, regardless of whether they are on a liquid or a gas.

The equations that fluid dynamics use to describe the motion of fluids are based on the hypothesis that fluid is *continuous*. For most common applications this hypothesis can be considered acceptable. If we consider that the free path of molecules in the gases is in the order of $10^{-7}m$, it is easy to expect that, only if you want a resolution lower than that scale, the hypothesis of the continuous is acceptable. For liquids, since the average free path of the molecules is about 3 orders of magnitude lower than that of the gas, the hypothesis of the continuous is even more applicable. During this Course we always assume that the hypothesis of continuity is acceptable, since the scales of interest are larger than those corresponding to the free path of the molecules.

The equations on which the fluid dynamics of reactive systems are based can be divided into two categories:

1. *Balance or conservation equations*: equations that express the conservation of total mass, momentum, and energy. Indeed, through the application of appropriate balances, it is possible to formulate similar equations, which more properly take the name of transport equations, for quantities that are not necessarily preserved (such as the masses of the individual species of a mixture). The most commonly used transport equations in fluid mechanics are the following:
 - individual chemical species (keep in mind that individual atomic elements are conserved and therefore it is possible to formulate equations of conservation);
 - entropy;
 - turbulent kinetic energy;
 - dissipation rate of turbulent kinetic energy;
 - Reynolds Stress Tensor (RST).
2. *Constitutive laws*: describing quantitatively the physical properties of fluids. The most important are: the Newton's law, the Fourier's law, and the Fick's law. Also the equations of state, which express the link between temperature, pressure, and density of a fluid, fall within this category.

The formulation of the transport equations of different quantities requires the use of some mathematical tools, which are not the expression or translation of any physical law, but which find their basis only on geometric aspects of fluid flow (kinematic theorems). The Leibnitz formula is particularly relevant to express the derivative of an integral with variable limits. *Leibnitz's theorem*, also known as the generalized transport theorem, is a generalization of particular importance to fluid mechanics. Often this theorem is also referred to as the *Reynolds transport theorem*, to underline its more general character than the Leibnitz formula.

In general, in order to be able to fully describe a reactive system, it is necessary to be able to manage the fluid thermodynamics. As everyone knows, thermodynamics is based on the concept of equilibrium states and assumes that any change in the fluid occurs in an extremely slow way, so that it goes through equilibrium states. However, the motion of a fluid in general is not necessarily slow and therefore the laws of classical thermodynamics could not work (in principle). However, what is done in practice is to assume that a state of equilibrium is reached locally so that the principles and laws of classical thermodynamics can be applied.

Of course, the transport equations can not be resolved analytically because of their complexity (in fact they are partial differential equations or PDEs). Numerical approaches must therefore be used, which are able to guarantee good accuracy and robustness, and which will be analyzed in more detail during this Course.

2 Transport Equations

In the following, we will introduce equations describing the behavior of multicomponent mixtures in a general vector form. The proposed equations can then be expressed in any reference system (e.g. Cartesian, cylindrical, or spherical) by using mathematical tools. This ensures that the equations have the same physical meaning, regardless of the reference system being considered. In general, in the field of computational fluid dynamics, it should be noted that non-orthogonal curvilinear coordinate systems are often used when very complex geometries have to be treated. So having the equations in a form independent of the coordinate system is of utmost importance.

2.1 Lagrangian vs Eulerian formulations

There are several approaches through which conservation equations (and more generally transport equations) can be obtained. First of all, it is appropriate to introduce the concept of *control system* and *control volume* on which the conservation equations are based. The term control system is used to uniquely identify a set of atoms and molecules (mass) that are a very precise part of the material being considered (not necessarily a fluid); such a set of atoms and molecules can interact with other control systems surrounding it, but without exchanging mass with them. A control volume, on the other hand, is a mass-independent geometric entity, which corresponds to a very precise volume of material, in principle variable over time. One aspect to be immediately clarified is that in principle the fluid dynamics equations should be formulated in terms of control system rather than control volumes. However, from a practical and applicative point of view it is much more convenient to have equations written on the control volumes and for this reason you have to be able to switch from one formulation to another. This possibility is guaranteed by the *Reynolds transport theorem* (or *Leibnitz theorem*) in the integral form. In general fluid mechanics prefers to work in terms of control volumes; structural mechanics in terms of control system.

There is a particular control volume of great importance, which at the same time also has the features of a control system and that is defined as the *material control volume*. With this term, we refer to the volume of fluid that contains always and only the entire mass of fluid that was present within it at the initial instant. The surface that delimits this volume is defined as *control surface* (CS). The material control volume is therefore a geometric entity, but defined on the mass. When talking about material control volume, the term Lagrangian is also used. Contrary to Lagrangian's point of view, there is the Eulerian's point of view, which takes into account a fixed volume control in space.

For the derivation of conservation and transport equations, there is no *a priori* better solution between the Lagrangian and the Eulerian points of view. The most appropriate choice depends on the specific features of the problem that you are considering. In principle, it is possible to choose volumes of any size and shape. Conceptually, the simplest possible description of the fluid is represented by the choice of a material control volume (which is therefore not fixed in space) that moves at the same velocity of the fluid. Though this is the conceptually more intuitive formulation, it is rarely used since its translation



Figure 1: Giuseppe Luigi Lagrange (1736-1813) and Leonhard Euler (1707-1783)

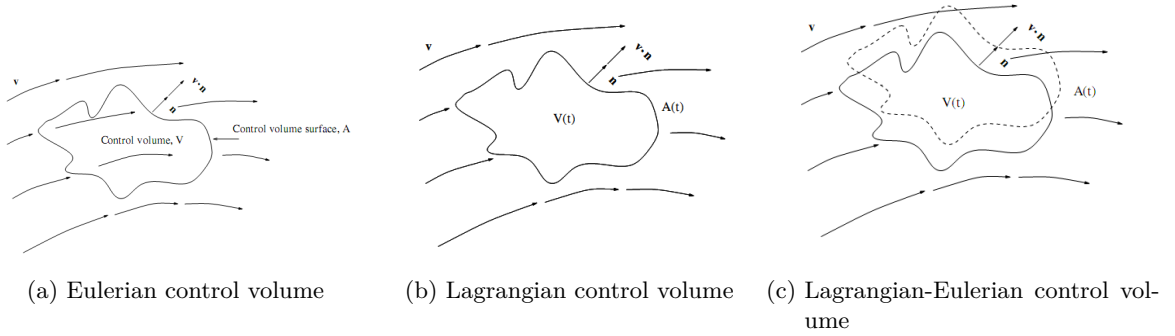


Figure 2: Examples of control volumes

is rather complex in mathematical terms, since control volume changes shape and size. It is easier to write equations using fixed control volumes (Eulerian), as shown in Figure 2, because in general this gives you the chance to get easier equations to deal with mathematically. However, keep in mind that the most general control volume is defined by surfaces that are not fixed in space and that do not necessarily move with the same velocity of the fluid, but at an arbitrary speed. This more general description takes the name of Eulerian-Lagrangian formulation (or generalized Lagrangian or Arbitrary Lagrangian-Eulerian (ALE)). This approach is, however, unused because it is too complex from the mathematical point of view.

Conservation and transport equations can be formulated in three equivalent main approaches:

1. Lagrangian formulation

A fluid particle that moves in space with a fixed reference system is taken into account. Then the balance equation is applied on the quantity of interest. This balance is based on the assumption that the value of any fluid property at the position $\mathbf{r}$ and at time t coincides with that of the infinitesimal particle occupying the position $\mathbf{r}$ at the instant t . Through the substantial derivative operator (defined below), one can move from this Lagrangian to the Eulerian formulation.

2. Eulerian formulation

An infinitesimal cube of fluid is always taken into account (fixed in space). By applying an appropriate balance, it is possible to obtain the conservation or transport equations of any quantity.

3. Macroscopic formulation

A finite size control volume is considered, which can be arbitrary in shape and can be fixed or moving. Applying the general balance equation, you get an equation in integral formulation. At this point, using the Leibnitz's theorem to define the rate of change and the Gauss theorem to transform the surface integrals into volume integrals, we get an equation that is similar to what we can obtained with either of the above approaches.

Let us spend some additional words on this last approach, which is of particular interest from the engineering point of view. The most general possible form of balance for any variable ψ on an arbitrary and finite size control volume (CV) has to account for accumulation, exchange of convective and diffusive nature, and a production term:

$$\left[\begin{array}{c} \text{accumulation} \\ \text{of } \psi \end{array} \right] = \left[\begin{array}{c} \text{net convective} \\ \text{flow rate of } \psi \end{array} \right] + \left[\begin{array}{c} \text{net diffusion} \\ \text{flow rate of } \psi \end{array} \right] + \left[\begin{array}{c} \text{net production} \\ \text{of } \psi \end{array} \right] \quad (2)$$

If we define V as the volume of the system, with A its surface, with $\hat{\mathbf{n}}$ the normal versor on this surface, directed to the outside, the mathematical translation of the balance written above is the following:

$$\frac{\partial}{\partial t} \int_V (\rho\psi) dV = - \int_A (\rho\mathbf{v}\psi) \cdot \hat{\mathbf{n}} dA - \int_A \mathbf{J} \cdot \hat{\mathbf{n}} dA + \int_V \Phi dV \quad (3)$$

In the above equation $\mathbf{v}$ indicates the velocity of the fluid, ρ its density, $\mathbf{J}$ is the diffusive flow of ψ and Φ is the production term (per unit of volume).

The derivative operation can be reversed with respect to the derivation and the surface integrals can be rewritten as volume integrals through the Gauss theorem (or Green theorem). This leads to the following result:

$$\int_V \left[\frac{\partial}{\partial t} (\rho\psi) + \nabla \cdot (\rho\mathbf{v}\psi) + \nabla \cdot \mathbf{J} - \Phi \right] dV = 0 \quad (4)$$

This equation must be valid for any macroscopic volume V and hence the function inside the integral must be identically equal to zero. Thus we have the equation we were looking for:

$$\frac{\partial}{\partial t} (\rho\psi) + \nabla \cdot (\rho\mathbf{v}\psi) = -\nabla \cdot \mathbf{J} + \Phi \quad (5)$$

Before moving to conservation and transport equations, it is important to define the concept of substantial derivative (or Lagrangian derivative) and finally introduce the transport theorem.

Substantial derivative (or Lagrangian derivative). Let's consider a generic scalar function $\psi = \psi(\mathbf{r}, t)$ (actually at this level it is not important if ψ is a vector or a scalar). The partial derivative with respect to the time $\frac{\partial \psi}{\partial t}$ is the derivative of ψ with respect to time maintaining the space variables constant: $\frac{\partial \psi}{\partial t} = \frac{\partial \psi}{\partial t} \Big|_{\mathbf{r}}$.

On the other hand, the total derivative is completely different. In particular, in Cartesian coordinates:

$$\frac{d\psi}{dt} = \frac{\partial \psi}{\partial t} \Big|_{x,y,z} + \frac{\partial \psi}{\partial x} \frac{dx}{dt} \Big|_{t,y,z} + \frac{\partial \psi}{\partial y} \frac{dy}{dt} \Big|_{t,x,z} + \frac{\partial \psi}{\partial z} \frac{dz}{dt} \Big|_{t,x,y} \quad (6)$$

In a more compact vector format (and independent of the coordinate system) you can write:

$$\frac{d\psi}{dt} = \frac{\partial \psi}{\partial t} \Big|_{x,y,z} + \mathbf{v}_S \cdot \nabla \psi \quad (7)$$

If the observer has a velocity exactly equal to that of the fluid $\mathbf{v}_S = \mathbf{v}$ the total derivative takes the name of material derivative or substantial derivative or Lagrangian derivative. In vector terms we can write:

$$\frac{D\psi}{Dt} = \frac{\partial \psi}{\partial t} \Big|_{x,y,z} + \mathbf{v} \cdot \nabla \psi \quad (8)$$

This derivative is of high interest because it is the point of view of an observer moving at the same velocity of fluid.

The transport theorem In order to identify the relationship between the Lagrangian and the Eulerian formulations we must be able to write the rate of variation of the integrals (of scalars or vectors) made on a moving material volume occupied by the fluid. Let us consider a material volume $V(t)$ delimited by an area $A(t)$ whose points move at the velocity of the fluid $\mathbf{v} = \mathbf{v}(\mathbf{r}, t)$. The substantial derivative of the integral of any function $f(\mathbf{r}, t)$ on the material control volume (which is moving) $V(t)$ is given by the following transport theorem:

$$\frac{D}{Dt} \int_{V(t)} f(\mathbf{r}, t) dV = \int_V \frac{\partial f(\mathbf{r}, t)}{\partial t} dV + \int_A f(\mathbf{r}, t) (\mathbf{v} \cdot \hat{\mathbf{n}}) dA \quad (9)$$

The volume V that appears to the second member is a fixed volume in the space, which coincides with the material volume $V(t)$ at the moment considered t . Similarly, surface A coincides with $A(t)$ at time t . The second term on the RHS represents the flux of the quantity f through the fixed surface of the control volume V . If we now apply the Gauss theorem, we have:

$$\int_A f(\mathbf{r}, t) (\mathbf{v} \cdot \hat{\mathbf{n}}) dA = \int_V \nabla \cdot (f\mathbf{v}) dV \quad (10)$$

The transport theorem can therefore be written in the following form, in which the LHS represent the Lagrangian term, while the RHS represents the Eulerian term:

$$\frac{D}{Dt} \int_{V(t)} f(\mathbf{r}, t) dV = \int_V \left[\frac{\partial f(\mathbf{r}, t)}{\partial t} + \nabla \cdot (f\mathbf{v}) \right] dV \quad (11)$$

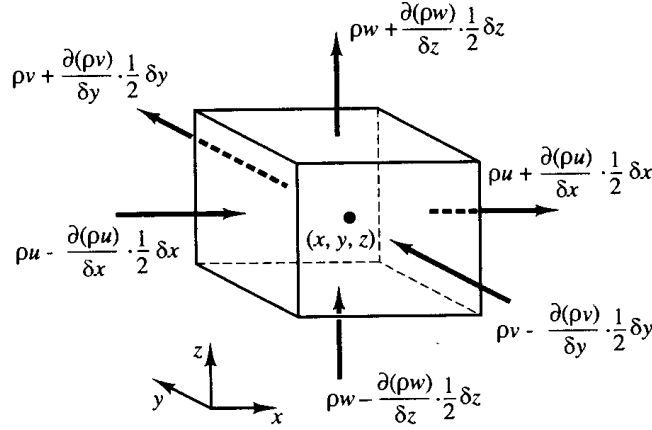


Figure 3: Eulerian control volume for continuity equation

3 Conservation of mass (continuity equation)

Let's start with the conservation equation of total mass, also known as *continuity equation*. Let's consider all three approaches that were previously reported.

3.1 Eulerian approach

Let us consider the infinitesimal cubic volume shown in Figure 3 and let us adopt a Cartesian reference system centered on the cube. The mass balance written on the cube obviously has the form that has already been discussed:

$$\left[\begin{array}{c} \text{accumulation} \\ \text{of mass} \end{array} \right] = \left[\begin{array}{c} \text{inlet mass} \\ \text{flow rate} \end{array} \right] - \left[\begin{array}{c} \text{outlet mass} \\ \text{flow rate} \end{array} \right] \quad (12)$$

Clearly, there is no term of production, being the balance made on the total mass of the system. The term of accumulation is simply equal to the derivative with respect to the time of the total mass contained in the volume and hence:

$$\frac{\partial m}{\partial t} = \frac{\partial}{\partial t} (\rho \delta V) = \frac{\partial}{\partial t} (\rho \delta x \delta y \delta z) = \frac{\partial \rho}{\partial t} \delta x \delta y \delta z \quad (13)$$

Exchange terms can be written by multiplying the fluxes across each cube face times the corresponding area. Since the only information available is the velocity value (and its components) at the center of the cube, it is necessary to make a Taylor series expansion (at the first order) in order to have the velocity of the fluid at the sides of the cube. In particular, for the two faces perpendicular to the x axis, we have the following two fluxes:

$$\rho v_x|_{x-\frac{1}{2}\delta x} \simeq \rho v_x - \frac{\partial (\rho v_x)}{\partial x} \frac{1}{2} \delta x$$

$$\rho v_x|_{x+\frac{1}{2}\delta x} \simeq \rho v_x + \frac{\partial (\rho v_x)}{\partial x} \frac{1}{2} \delta x$$

The net mass flow outward from the perpendicular to the x axis is given by:

$$\left[\rho v_x|_{x+\frac{1}{2}\delta x} - \rho v_x|_{x-\frac{1}{2}\delta x} \right] \delta y \delta z \simeq \left[\rho v_x + \frac{\partial (\rho v_x)}{\partial x} \frac{1}{2} \delta x - \rho v_x + \frac{\partial (\rho v_x)}{\partial x} \frac{1}{2} \delta x \right] \delta y \delta z = \frac{\partial (\rho v_x)}{\partial x} \delta x \delta y \delta z$$

Likewise, the massive net flow rates can be estimated through the remaining two pairs of faces:

$$\left[\rho v_y|_{y+\frac{1}{2}\delta y} - \rho v_y|_{y-\frac{1}{2}\delta y} \right] \delta x \delta z \simeq \left[\rho v_y + \frac{\partial(\rho v_y)}{\partial y} \frac{1}{2} \delta y - \rho v_y + \frac{\partial(\rho v_y)}{\partial y} \frac{1}{2} \delta y \right] \delta x \delta z = \frac{\partial(\rho v_y)}{\partial y} \delta x \delta y \delta z$$

$$\left[\rho v_z|_{z+\frac{1}{2}\delta z} - \rho v_z|_{z-\frac{1}{2}\delta z} \right] \delta x \delta y \simeq \left[\rho v_z + \frac{\partial(\rho v_z)}{\partial z} \frac{1}{2} \delta z - \rho v_z + \frac{\partial(\rho v_z)}{\partial z} \frac{1}{2} \delta z \right] \delta x \delta y = \frac{\partial(\rho v_z)}{\partial z} \delta x \delta y \delta z$$

Then the total terms of exchange can be written as:

$$IN - OUT = - \left[\frac{\partial(\rho v_x)}{\partial x} + \frac{\partial(\rho v_y)}{\partial y} + \frac{\partial(\rho v_z)}{\partial z} \right] \delta x \delta y \delta z = -\nabla \cdot (\rho \mathbf{v}) \delta x \delta y \delta z$$

In the equation above, the divergence $\nabla \cdot$ of the vector $\rho \mathbf{v}$ has been introduced in order to arrive to a more compact equation. Note that the divergence of a vector $\boldsymbol{\psi}$ is equal to the following scalar quantity:

$$\nabla \cdot \boldsymbol{\psi} = \frac{\partial \psi_x}{\partial x} + \frac{\partial \psi_y}{\partial y} + \frac{\partial \psi_z}{\partial z}$$

The global balance is therefore:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0 \quad (14)$$

Observation I. The above equation, although obtained for a Cartesian coordinate system, is actually valid also in a different reference system (e.g. cylindrical or spherical coordinates). For example, in cylindrical coordinates, the divergence operator has the following form:

$$\nabla(\varphi) = \frac{1}{r} \frac{\partial}{\partial r} (r \varphi_r) + \frac{1}{r} \frac{\partial \varphi_\theta}{\partial \theta} + \frac{\partial \varphi_z}{\partial z} \quad (15)$$

Observation II. Often the continuity equation is written in more useful and convenient forms. Let's remember the concept of a substantial D/Dt derivative whose physical meaning is that of a temporal derivative made by an observer moving at the same fluid velocity:

$$\frac{D(\cdot)}{Dt} = \frac{\partial(\cdot)}{\partial t} + \mathbf{v} \cdot \nabla(\cdot) \quad (16)$$

Since the $\nabla \cdot$ and ∇ operators are linear, i.e.:

$$\nabla \cdot (a \mathbf{b}) = a \nabla \cdot \mathbf{b} + \mathbf{b} \cdot \nabla a \quad (17)$$

the continuity equation can then be rewritten in this new equivalent form in which the substantial derivative appears explicitly:

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot (\rho \mathbf{v}) = \rho \nabla \cdot \mathbf{v} + \mathbf{v} \cdot \nabla \rho = \frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{v} = 0 \quad (18)$$

So the continuity equation assumes the following form:

$$\frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{v} = 0 \quad (19)$$

Observation III. Note that it is not possible to write a similar equation in molar terms, basically because the reactions take place in general with variation in the number of moles, and therefore there is a production term linked to chemical reactions (which, of course, in mass terms is null). Also, we have also to account for that the molar average velocity $\mathbf{v}^*$ is generally different from the velocity $\mathbf{v}$ defined on a massive basis:

$$\mathbf{v}^* = \frac{\sum_{j=1}^{NC} C_j \mathbf{v}_j}{\sum_{j=1}^{NC} C_j} = \sum_{j=1}^{NC} x_j \mathbf{v}_j \neq \sum_{j=1}^{NC} \omega_j \mathbf{v}_j = \mathbf{v} \quad (20)$$

The equation of continuity in molar terms can be written as:

$$\frac{\partial C_{tot}}{\partial t} + \nabla \cdot (C_{tot} \mathbf{v}^*) = \sum_{j=1}^{NC} \tilde{R}_j \quad (21)$$

In the equation above, $\tilde{R}_j$ indicates the molar formation rate of species j . The physical meaning of this equation is, however, more elusive than the corresponding equation written in massive terms, just because v^* is not the real fluid mechanics velocity.

3.2 Macroscopic approach

Consider a macroscopic volume V at the given time t , through whose surface A mass exchanges take place. The global mass balance can be written by imposing that the mass accumulation within the volume equals the difference between the incoming mass flow and the outgoing mass:

$$\frac{dm}{dt} = \dot{m}_{in} - \dot{m}_{out} \quad (22)$$

The mass contained within the volume can of course be written as:

$$m = \int_V \rho dV \quad (23)$$

Likewise, massive flows can be expressed through the incoming and outgoing mass flows through surface A :

$$\dot{m}_{in} - \dot{m}_{out} = - \int_A \rho \mathbf{v} \cdot \hat{\mathbf{n}} dA \quad (24)$$

The balance equation therefore becomes the following:

$$\frac{\partial}{\partial t} \left(\int_V \rho dV \right) = - \int_A \rho \mathbf{v} \cdot \hat{\mathbf{n}} dA \quad (25)$$

Thanks to the Gauss theorem, the surface integral can be written as a volume integral:

$$\frac{\partial}{\partial t} \left(\int_V \rho dV \right) = - \int_V \nabla \cdot (\rho \mathbf{v}) dV \quad (26)$$

Thus:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0 \quad (27)$$

Similarly, you can get the equation of continuity in molar terms (already written above):

$$\frac{\partial C_{tot}}{\partial t} + \nabla \cdot (C_{tot} \mathbf{v}^*) = \sum_{j=1}^{NC} \tilde{R}_j \quad (28)$$

3.3 Lagrangian approach

It is also possible to use the Lagrangian approach that has been described above to determine the total mass conservation equation. The total mass contained within a finite volume control volume $V(t)$ is given by the following expression:

$$m = \int_{V(t)} \rho dV \quad (29)$$

The mass conservation principle can be formulated for this system characterized by a volume $V(t)$ that moves in space and delimited by surface $A(t)$, taking into account that the total mass within the material control volume can not change, since total mass is not produced or destroyed (unless nuclear reactions occur):

$$\frac{Dm}{Dt} = \frac{D}{Dt} \int_{V(t)} \rho dV = 0 \quad (30)$$

The fundamental transport theorem allows to transform the above equation into an Eulerian formulation:

$$\frac{D}{Dt} \int_{V(t)} \rho dV = \int_V \left[\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) \right] dV = 0 \quad (31)$$

The above equation must be satisfied for any volume V , so the expression inside the integral must be equal to zero. Thus, we arrive to the continuity equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0 \quad (32)$$

4 Conservation of chemical species

The species conservation equations can be written in either massive or molar terms. Generally, in the field of fluid dynamics, the massive formulation is preferred to exploit some properties that are not valid in molar terms. However, it is easy to move from one formulation to another. Before introducing the transport equations, some important definitions should be introduced.

The individual chemical species in a mixture move at different speeds. In particular, $\mathbf{v}_j$ is the velocity of species j in respect of the reference system. For a mixture of NC species, fluiddynamic velocity (or mass average speed) is defined as (already mentioned

in the previous paragraph):

$$\mathbf{v} = \frac{\sum_{j=1}^{NC} \rho_j \mathbf{v}_j}{\sum_{j=1}^{NC} \rho_j} = \frac{\sum_{j=1}^{NC} \rho_j \mathbf{v}_j}{\rho} = \sum_{j=1}^{NC} \omega_j \mathbf{v}_j \quad (33)$$

where ρ is the density of the mixture and ω_j is the mass fraction of species j .

It is then important to know not only the absolute velocity of a species, but also its relative velocity with respect to the fluid velocity. This leads to the definition of *diffusive velocity*:

$$\mathbf{v}_{j,d} = \mathbf{v}_j - \mathbf{v} \quad (34)$$

Now we can define the mass diffusion fluxes of species. The total massive flow of species j is a vector quantity that indicates the mass of the species that crosses the surface unit in the unit of time. His expression is:

$$\dot{\mathbf{m}}_j = \rho_j \mathbf{v}_j \quad (35)$$

The diffusive (or relative) mass flow is instead defined on diffusive velocity:

$$\mathbf{j}_j = \rho_j (\mathbf{v}_j - \mathbf{v}) = \rho_j \mathbf{v}_{j,d} \quad (36)$$

Convective fluxes on the contrary are defined as follows:

$$\rho_j \mathbf{v} = \dot{\mathbf{m}}_j - \mathbf{j}_j \quad (37)$$

4.1 Eulerian approach

Let us consider the usual infinitesimal volume in a Cartesian reference system (shown in Figure 3). The rate of change of the total mass of the species j within the volume is given by:

$$\frac{\partial m_j}{\partial t} = \frac{\partial}{\partial t} (\rho_j \delta x \delta y \delta z) = \frac{\partial \rho_j}{\partial t} \delta x \delta y \delta z \quad (38)$$

The balance on the individual species j requires that the mass of species j contained in the volume can change only by the convective and diffusive exchange of matter and chemical reactions:

$$\frac{\partial m_j}{\partial t} = \left[\frac{\partial m_j}{\partial t} \right]_{conv} + \left[\frac{\partial m_j}{\partial t} \right]_{diff} + \left[\frac{\partial m_j}{\partial t} \right]_{reaz} \quad (39)$$

Convective exchange terms can be written as it was done for the continuity equation. First of all, we must determine the material flows of the species j on the 6 sides of the cube and from these it is possible to derive the corresponding flows. If we focus attention on the x axis perpendicular faces, a Taylor series expansion around the center of the reference system (and hence at the center of the cube) allows us writing the fluxes of species j along the two faces under examination:

$$\rho_j v_x|_{x-\frac{1}{2}\delta x} \simeq \rho_j v_x - \frac{\partial (\rho_j v_x)}{\partial x} \frac{1}{2} \delta x$$

$$\rho_j v_x|_{x+\frac{1}{2}\delta x} \simeq \rho_j v_x + \frac{\partial(\rho_j v_x)}{\partial x} \frac{1}{2} \delta x$$

The net flow rate exiting through the two faces perpendicular to the x direction is:

$$\begin{aligned} & \left[\rho_j v_x|_{x+\frac{1}{2}\delta x} - \rho_j v_x|_{x-\frac{1}{2}\delta x} \right] \delta y \delta z \simeq \\ & \left[\rho_j v_x + \frac{\partial(\rho_j v_x)}{\partial x} \frac{1}{2} \delta x - \rho_j v_x + \frac{\partial(\rho_j v_x)}{\partial x} \frac{1}{2} \delta x \right] \delta y \delta z = \frac{\partial(\rho_j v_x)}{\partial x} \delta x \delta y \delta z \end{aligned}$$

Similarly, for remaining faces:

$$\begin{aligned} & \left[\rho_j v_y|_{y+\frac{1}{2}\delta y} - \rho_j v_y|_{y-\frac{1}{2}\delta y} \right] \delta x \delta z \simeq \\ & \left[\rho_j v_y + \frac{\partial(\rho_j v_y)}{\partial y} \frac{1}{2} \delta y - \rho_j v_y + \frac{\partial(\rho_j v_y)}{\partial y} \frac{1}{2} \delta y \right] \delta x \delta z = \frac{\partial(\rho_j v_y)}{\partial y} \delta x \delta y \delta z \end{aligned}$$

$$\begin{aligned} & \left[\rho_j v_z|_{z+\frac{1}{2}\delta z} - \rho_j v_z|_{z-\frac{1}{2}\delta z} \right] \delta x \delta y \simeq \\ & \left[\rho_j v_z + \frac{\partial(\rho_j v_z)}{\partial z} \frac{1}{2} \delta z - \rho_j v_z + \frac{\partial(\rho_j v_z)}{\partial z} \frac{1}{2} \delta z \right] \delta x \delta y = \frac{\partial(\rho_j v_z)}{\partial z} \delta x \delta y \delta z \end{aligned}$$

Thus:

$$\left[\frac{\partial m_j}{\partial t} \right]_{conv} = - \left[\frac{\partial(\rho_j v_x)}{\partial x} + \frac{\partial(\rho_j v_y)}{\partial y} + \frac{\partial(\rho_j v_z)}{\partial z} \right] \delta x \delta y \delta z = -\nabla \cdot (\rho_j \mathbf{v}) \delta x \delta y \delta z$$

Pay particular attention to the fact that convective velocity $\mathbf{v}$ has been used. The diffusive terms can be treated in a similar way, while substituting the convection flow $\rho_j \mathbf{v}$ the diffusive flow $\mathbf{j}_j$. So without getting into the details, we have:

$$\left[\frac{\partial m_j}{\partial t} \right]_{diff} = - \left[\frac{\partial(j_{j,x})}{\partial x} + \frac{\partial(j_{j,y})}{\partial y} + \frac{\partial(j_{j,z})}{\partial z} \right] \delta x \delta y \delta z = -(\nabla \cdot \mathbf{j}_j) \delta x \delta y \delta z$$

The reaction term can be easily written, since it is local

$$\left[\frac{\partial m_j}{\partial t} \right]_{reazione} = R_j \delta x \delta y \delta z \quad (40)$$

Then, the transport equation becomes:

$$\frac{\partial \rho_j}{\partial t} + \nabla \cdot (\rho_j \mathbf{v}) = -\nabla \cdot \mathbf{j}_j + R_j \quad (41)$$

Or, introducing the total mass flow rate:

$$\frac{\partial \rho_j}{\partial t} + \nabla \cdot (\rho_j \mathbf{v}_j) = R_j \quad (42)$$

Observation I. The above equation is valid in any geometry and reference system. The sum of the conservation equations for all NC species gives rise to the equation of continuity, which had been obtained independently:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0 \quad (43)$$

This result is easy to demonstrate since the sum of all diffusive flows is by definition equal to zero:

$$\sum_{j=1}^{NC} \mathbf{j}_j = \mathbf{0} \quad (44)$$

and, thanks to the law of conservation of the mass, the sum of the formation rates of all the chemical species is also identically equal to zero:

$$\sum_{j=1}^{NC} R_j = 0 \quad (45)$$

Observation II. The transport equation of the individual species can also be written in molar terms. Just divide the transport equation of species j by the molecular weight of the same M_j :

$$\frac{\partial}{\partial t} \left(\frac{\rho_j}{M_j} \right) + \nabla \cdot \left(\frac{\rho_j}{M_j} \mathbf{v}_j \right) = \frac{R_j}{M_j} \quad (46)$$

Taking into account that the ratio between ρ_j and M_j is by definition the concentration of species j , it has:

$$\frac{\partial C_j}{\partial t} + \nabla \cdot (C_j \mathbf{v}_j) = \tilde{R}_j \quad (47)$$

Observation III. Also the transport equation of the individual species is often written in a different form, introducing the substantial derivative. First, keep in mind that $\rho_j = \rho \omega_j$. Let us therefore consider the first member of the transport equation (41):

$$\frac{\partial \rho_j}{\partial t} + \nabla \cdot (\rho_j \mathbf{v}) = \omega_j \frac{\partial \rho}{\partial t} + \rho \frac{\partial \omega_j}{\partial t} + \nabla \cdot (\rho \omega_j \mathbf{v}) = \omega_j \frac{\partial \rho}{\partial t} + \rho \frac{\partial \omega_j}{\partial t} + \rho \mathbf{v} \cdot \nabla \omega_j + \omega_j \nabla \cdot (\rho \mathbf{v}) \quad (48)$$

$$\omega_j \left[\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) \right] + \rho \left[\frac{\partial \omega_j}{\partial t} + \mathbf{v} \cdot \nabla \omega_j \right] = \rho \left[\frac{\partial \omega_j}{\partial t} + \mathbf{v} \cdot \nabla \omega_j \right] = \rho \frac{D\omega_j}{Dt} \quad (49)$$

Thus:

$$\rho \frac{D\omega_j}{Dt} = -\nabla \cdot \mathbf{j}_j + R_j \quad (50)$$

If you want to introduce a similar equation in molar terms, difficulties arise from the definition of the substantial derivative to be constructed with velocity $\mathbf{v}^*$. Only if we assume that the density and total concentrations are little variable and that the velocity $\mathbf{v}^*$ can be confused with $\mathbf{v}$ we can write:

$$C_{tot} \frac{Dx_j}{Dt} \simeq -\nabla \cdot \mathbf{J}_j + \tilde{R}_j \quad (51)$$

In the above equation, the molar diffusive flow was introduced:

$$\mathbf{J}_j = C_j (\mathbf{v}_j - \mathbf{v}^*) \quad (52)$$

4.2 Macroscopic approach

The macroscopic approach assumes that a volume of finite size V is taken into consideration at the time t , delimited by the control surface A . The macroscopic conservation balance of species j in mass terms can be formulated as:

$$\left[\begin{array}{c} \text{accumulation of} \\ \text{mass of } j \end{array} \right] = \left[\begin{array}{c} \text{net convective} \\ \text{mass flow rate of } j \end{array} \right] + \left[\begin{array}{c} \text{net diffusion} \\ \text{flow rate of } j \end{array} \right] + \left[\begin{array}{c} \text{net production} \\ \text{of } j \end{array} \right] \quad (53)$$

This means:

$$\frac{\partial m_j}{\partial t} = [\dot{m}_j]_{conv} + [\dot{m}_j]_{diff} + [\dot{m}_j]_{reaz} \quad (54)$$

We can write the mass of species j within the volume as a function of ρ_j :

$$m_j = \int_V \rho_j dV \quad (55)$$

Net mass flow rate due to convective exchange is given by the surface integrale of convective fluxes of j thorough surface A :

$$[\dot{m}_j]_{conv} = - \int_A \rho_j \mathbf{v} \cdot \hat{\mathbf{n}} dA \quad (56)$$

We can proceed similarly for diffusion:

$$[\dot{m}_j]_{diff} = - \int_A \mathbf{j}_j \cdot \hat{\mathbf{n}} dA \quad (57)$$

The reaction term, being local, can be written as a volume integral accounting for the formation rate R_j :

$$[\dot{m}_j]_{reaz} = \int_V R_j dV \quad (58)$$

Thus:

$$\frac{\partial}{\partial t} \left(\int_V \rho_j dV \right) = - \int_A \rho_j \mathbf{v} \cdot \hat{\mathbf{n}} dA - \int_A \mathbf{j}_j \cdot \hat{\mathbf{n}} dA + \int_V R_j dV \quad (59)$$

Thanks to the Gauss theorem, the surface integral can be written as a volume integral:

$$\int_V \left[\frac{\partial \rho_j}{\partial t} + \nabla \cdot (\rho_j \mathbf{v}) + \nabla \cdot (\mathbf{j}_j) - R_j \right] dV = 0 \quad (60)$$

The above equation must be satisfied for any volume V , so the expression inside the integral must be equal to zero. Thus, we arrive to the equation we were looking for:

$$\frac{\partial \rho_j}{\partial t} + \nabla \cdot (\rho_j \mathbf{v}) = - \nabla \cdot \mathbf{j}_j + R_j \quad (61)$$

4.3 Lagrangian approach

For a single species in a mixture, the conservation principle on the mass of the individual species postulates that the rate of change of mass of species j equals the rate at which the mass of the same species is produced for chemical reactions. The mass of the species j in the finite material control volume $V_j(t)$ is given by the following expression:

$$m_j = \int_{V_j(t)} \rho_j dV \quad (62)$$

If $V_j(t)$ is the volume occupied by species j and R_j is the formation rate of the same species, we have:

$$\frac{Dm}{Dt} = \frac{D}{Dt} \int_{V_j(t)} \rho_j dV = \int_{V_j(t)} R_j dV \quad (63)$$

We can now apply the transport theorem for moving the formulation from Lagrangian point of view to the Eulerian point of view:

$$\frac{D}{Dt} \int_{V_j(t)} \rho_j dV = \int_{V_j} \left[\frac{\partial \rho_j}{\partial t} + \nabla \cdot (\rho_j \mathbf{v}_j) \right] dV \quad (64)$$

Then, accounting for the reaction term:

$$\int_{V_j} \left[\frac{\partial \rho_j}{\partial t} + \nabla \cdot (\rho_j \mathbf{v}_j) - R_j \right] dV = 0 \quad (65)$$

The above equation must be valid for a generic macroscopic volume V_j and this implies that the integral function is identically equal to zero. The transport equation of a chemical species (in conservative form) is thus obtained:

$$\frac{\partial \rho_j}{\partial t} + \nabla \cdot (\rho_j \mathbf{v}_j) = R_j \quad (66)$$

In general, it is more convenient to rewrite the above equation by distinguishing convective from diffusion fluxes:

$$\frac{\partial \rho_j}{\partial t} + \nabla \cdot (\rho_j \mathbf{v}) = -\nabla \cdot \mathbf{j}_j + R_j \quad (67)$$

5 Conservation of momentum

5.1 Molecular stress tensor

Let us consider a cubic fluid element (as usual), characterized by a velocity $\mathbf{v}$ oriented in any direction. The center of the cube occupies the position (x, y, z) . At time t we imagine we *slice* the cube exactly in the center using a plane perpendicular to any of the three axes. We begin to examine the situation corresponding to a *slicing* obtained through a plane perpendicular to the x axis, as shown in Figure 4. We can look for the force per unit of surface π_x that must be applied to the face obtained after the slicing to compensate exactly what was exerted by the fluid that was removed. This is obviously a vector that can be oriented in any direction. We can always split this vector in a component perpendicular to the surface, which we call $p\delta_x$ (where δ_x is the versor parallel to the x

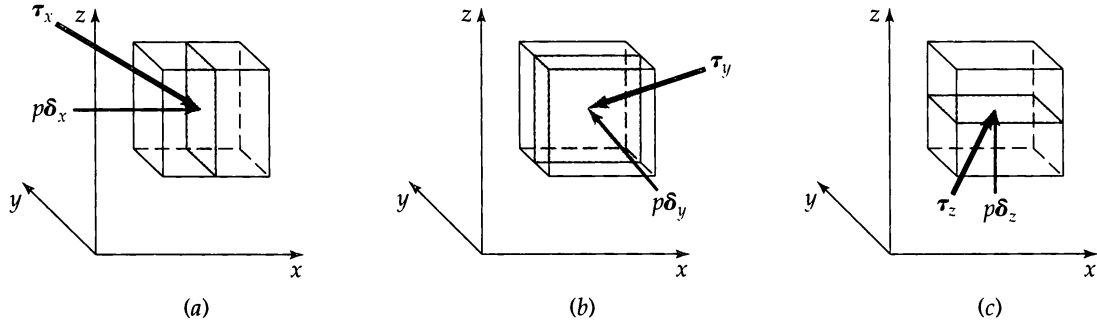


Figure 4: Pressure and viscous forces acting over the faces perpendicular to the three main axes of the Cartesian system for a cubic volume

axis and oriented in the same direction of this and p is fluid pressure). The remaining part, which we indicate with τ_x , can have any direction too. Obviously the same can be repeated for y and z directions as well. In particular, the forces associated with the pressure are perpendicular to the surface itself. The other contribution is generally more complex to deal with, because *a priori* it is neither parallel nor perpendicular to the surface, but has any direction. However, each of the three vector τ_x , τ_y and τ_z can also be decomposed into the three components parallel to the axes .

We can write for π_x :

$$\pi_x = p \delta_x + \tau_x \quad (68)$$

Similarly, for the remaining 2 directions:

$$\pi_y = p \delta_y + \tau_y \quad (69)$$

$$\pi_z = p \delta_z + \tau_z \quad (70)$$

We can adopt a more compact tensorial formulation, defining the hydrostatic stress tensor $p\delta$, the viscous stress tensor $\underline{\tau}$ and the molecular stress tensor $\underline{\pi}$:

$$\underline{\pi} = p\delta + \underline{\tau} \quad (71)$$

In particular, with the symbol π_{ij} we mean: the component along the direction j of the force perpendicular to a unit element with surface perpendicular to the direction i (with the convention that the fluid in the region of smaller x_i is smaller force on regions with greater x_i). The $\pi_{xx} = p + \tau_{xx}$, $\pi_{yy} = p + \tau_{yy}$ and $\pi_{zz} = p + \tau_{zz}$ stresses are called *normal stresses*, obviously because they all correspond to all and only the stresses acting in the normal direction to surfaces perpendicular to directions x , y and z , respectively. The remaining efforts $\pi_{xy} = \tau_{xy}$, $\pi_{xz} = \tau_{xz}$, ... are called *shear stresses*. The matrix of molecular stresses is summarized in Table 1.

5.2 Convective flux tensor

The momentum per unit of volume associated to the cubic volume under investigation is equal to $\rho \mathbf{v}$. If we now calculate the flux of momentum per unit of volume through the surface resulting from the *slicing*, we get $\rho \mathbf{v} v_x$ (be aware that this is measured from regions at lower x to regions at higher x). If we look at its dimensions, we can easily

Direction normal to the surface	Force per unit of surface	Force components		
		x component	y component	z component
x	$\boldsymbol{\pi}_x = p \boldsymbol{\delta}_x + \boldsymbol{\tau}_x$	$\pi_{xx} = p + \tau_{xx}$	$\pi_{xy} = \tau_{xy}$	$\pi_{xz} = \tau_{xz}$
y	$\boldsymbol{\pi}_y = p \boldsymbol{\delta}_y + \boldsymbol{\tau}_y$	$\pi_{yx} = \tau_{yx}$	$\pi_{yy} = p + \tau_{yy}$	$\pi_{yz} = \tau_{yz}$
z	$\boldsymbol{\pi}_z = p \boldsymbol{\delta}_z + \boldsymbol{\tau}_z$	$\pi_{zx} = \tau_{zx}$	$\pi_{zy} = \tau_{zy}$	$\pi_{zz} = p + \tau_{zz}$

Table 1: Molecular stress tensor

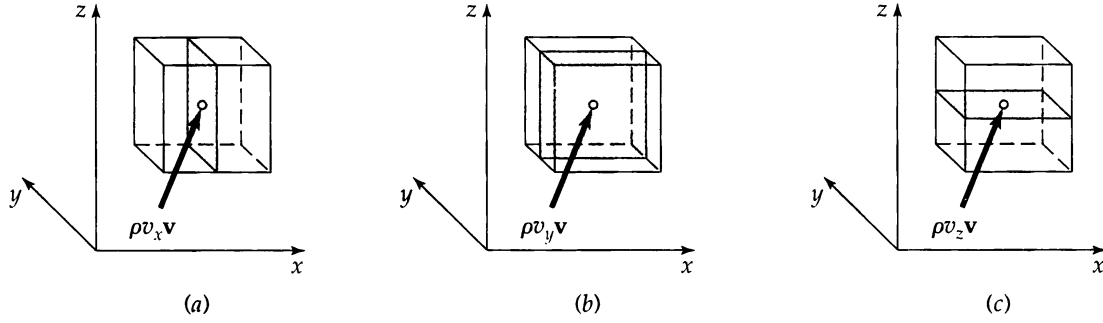


Figure 5: Convective fluxes of momentum through the faces perpendicular to the three main axes

verify that it is a force per unit of surface, that is a stress or a pressure. Similarly, flows through normal surfaces to the other two directions will be $\rho \mathbf{v} v_y$ and $\rho \mathbf{v} v_z$, respectively (Figure 5). Therefore, the three vectors $\rho \mathbf{v} v_x$, $\rho \mathbf{v} v_y$ and $\rho \mathbf{v} v_z$ describe the fluxes of momentum per unit of volume through the surfaces perpendicular to the axes x, y and z , respectively. Obviously these are vectors and therefore each of them can be decomposed into its components parallel to the three axes, which are summarized in Table 2. The physical interpretation of such variables is fairly clear: for example, $\rho v_x v_y$ is the convective flux of the y component of momentum across a surface perpendicular to the x axis. More concisely, the set of these nine scalar components can be written as a tensor:

$$\rho \mathbf{v} \mathbf{v} = \left(\sum_i \delta_i \rho v_i \right) \mathbf{v} = \left(\sum_i \delta_i \rho v_i \right) \left(\sum_j \delta_j v_j \right) = \sum_i \sum_j \delta_i \delta_j \rho v_i v_j \quad (72)$$

This tensor is called *convective flux tensor* of momentum.

5.3 Combined flux tensor

Let us compare the units of the molecular and convective tensors defined in the previous sections. We easily realize that the units are exactly the same, i.e. those of a force per unit of surface or, if you like, of a pressure. Thanks to this information we can provide a physical interpretation of the stress tensors, which is alternative to what we have already used. In particular, if we refer to π_{ij} we can state that this represents the flux of component j of momentum per unit of volume in the i direction. This new interpretation is particularly useful because it directly compares the tensor of molecular stresses to the convective flux tensor $\rho \mathbf{v} \mathbf{v}$. In particular, we can collect all these stresses in a single tensor Φ , generally defined as a *combined tensor of momentum fluxes*:

Direction normal to the surface	Flux of momentum per unit of volume	Momentum flux components		
		x component	y component	z component
x	$\rho \mathbf{v} v_x$	$\rho v_x v_x$	$\rho v_x v_y$	$\rho v_x v_z$
y	$\rho \mathbf{v} v_y$	$\rho v_y v_x$	$\rho v_y v_y$	$\rho v_y v_z$
z	$\rho \mathbf{v} v_z$	$\rho v_z v_x$	$\rho v_z v_y$	$\rho v_z v_z$

Table 2: Convective flux tensor of momentum

Tensor	Physical meaning
$\rho \mathbf{v} \mathbf{v}$	convective fluxes
$\underline{\boldsymbol{\tau}}$	viscous fluxes
$\underline{\boldsymbol{\pi}} = p \underline{\boldsymbol{\delta}} + \underline{\boldsymbol{\tau}}$	molecular fluxes
$\underline{\boldsymbol{\Phi}} = \underline{\boldsymbol{\pi}} + \rho \mathbf{v} \mathbf{v}$	combined fluxes

Table 3: Momentum fluxes

$$\underline{\boldsymbol{\Phi}} = \underline{\boldsymbol{\pi}} + \rho \mathbf{v} \mathbf{v} = p \underline{\boldsymbol{\delta}} + \underline{\boldsymbol{\tau}} + \rho \mathbf{v} \mathbf{v} \quad (73)$$

Overall, the momentum flux is generally associated with two main contributions: one directly related to the stresses that the fluid is able to carry out and one directly related to the only convective (inertial) flow.

This tensor results from the sum of three tensors. Of these, $p \underline{\boldsymbol{\delta}}$ does not depend on velocity and therefore it is never zero. The other two are the function of fluid velocity. In light of what has been reported, Φ_{ij} can be considered as the total flux of component j of momentum per unit of volume through a surface perpendicular to the axis i (due to convective and molecular mechanisms).

In the Table 3 all the tensors are summarized. For the sake of clarity, since they may be useful in the following, the matrix expression of such tensors is also explicitly reported:

$$\begin{bmatrix} \Phi_{xx} & \Phi_{xy} & \Phi_{xz} \\ \Phi_{yx} & \Phi_{yy} & \Phi_{yz} \\ \Phi_{zx} & \Phi_{zy} & \Phi_{zz} \end{bmatrix} = \begin{bmatrix} p & 0 & 0 \\ 0 & p & 0 \\ 0 & 0 & p \end{bmatrix} + \begin{bmatrix} \tau_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \tau_{yy} & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \tau_{zz} \end{bmatrix} + \rho \begin{bmatrix} v_x v_x & v_x v_y & v_x v_z \\ v_y v_x & v_y v_y & v_y v_z \\ v_z v_x & v_z v_y & v_z v_z \end{bmatrix} \quad (74)$$

Momentum balance

Now we have all the elements to get the equation of balance of momentum. Unlike the conservation equation of the total mass, since the momentum is a vector, the corresponding conservation equation will also be in a vector form and therefore it is expected to be more complex. It is more convenient to work on a single component per time. In this way, three scalar conservation equations will be obtained, one for each of the three components of momentum. Of course we begin with the component parallel to the x axis.

We must therefore write a conservation balance of component x of momentum on the cubic volume. As usual, in the most general form possible you have:

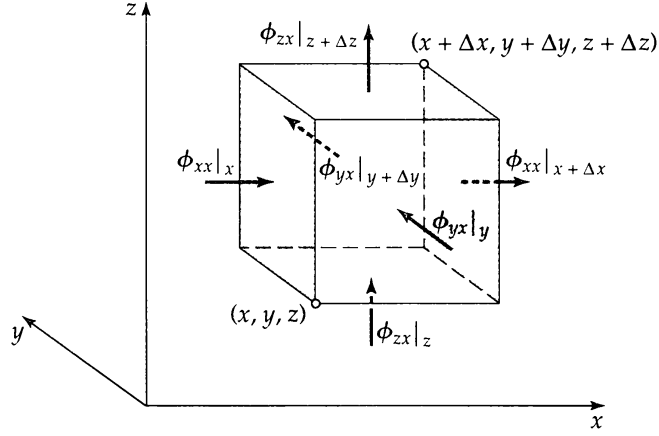


Figure 6: Control volume for momentum equation

$$\left[\begin{array}{c} \text{accumulation} \\ \text{of momentum} \end{array} \right] = \left[\begin{array}{c} \text{inlet flow rate} \\ \text{of momentum} \end{array} \right] - \left[\begin{array}{c} \text{outlet flow rate} \\ \text{of momentum} \end{array} \right] + \left[\begin{array}{c} \text{external} \\ \text{forces} \end{array} \right] \quad (75)$$

The accumulation term is: $\frac{\partial(\rho v_x)}{\partial t} \delta x \delta y \delta z$.

The input and output terms, on the other hand, require the knowledge of the total fluxes of momentum over the six faces. Since the volume cube is small, these flows can be determined by Taylor's expansion stopped at the first order. We have to be careful: the momentum flux is a vector and it is thus described by three components. We are now interested only in the component parallel to the x axis for each face. Therefore, for faces perpendicular to the x axis we only need the component Φ_{xx} , for the faces perpendicular to the y axis only the components Φ_{yx} and for z only Φ_{zx} (remember that in Φ_{ij} , j indicates the vector component and i is the normal to the surface). So, for faces perpendicular to the x axis:

$$\Phi_{xx}|_{x-\frac{1}{2}\delta x} \simeq \Phi_{xx} - \frac{\partial(\Phi_{xx})}{\partial x} \frac{1}{2} \delta x \quad (76)$$

$$\Phi_{xx}|_{x+\frac{1}{2}\delta x} \simeq \Phi_{xx} + \frac{\partial(\Phi_{xx})}{\partial x} \frac{1}{2} \delta x \quad (77)$$

For faces perpendicular to the y axis:

$$\Phi_{yx}|_{y-\frac{1}{2}\delta y} \simeq \Phi_{yx} - \frac{\partial(\Phi_{yx})}{\partial y} \frac{1}{2} \delta y \quad (78)$$

$$\Phi_{yx}|_{y+\frac{1}{2}\delta y} \simeq \Phi_{yx} + \frac{\partial(\Phi_{yx})}{\partial y} \frac{1}{2} \delta y \quad (79)$$

For faces perpendicular to the z axis:

$$\Phi_{zx}|_{z-\frac{1}{2}\delta z} \simeq \Phi_{zx} - \frac{\partial(\Phi_{zx})}{\partial z} \frac{1}{2} \delta z \quad (80)$$

$$\Phi_{yzx}|_{z+\frac{1}{2}\delta z} \simeq \Phi_{yzx} + \frac{\partial(\Phi_{yzx})}{\partial z} \frac{1}{2} \delta z \quad (81)$$

Thus, net fluxes along the three directions are given by:

$$\Phi_{xx}|_{x+\frac{1}{2}\delta x} - \Phi_{xx}|_{x-\frac{1}{2}\delta x} = \frac{\partial(\Phi_{xx})}{\partial x}\delta x \quad (82)$$

$$\Phi_{yx}|_{y+\frac{1}{2}\delta y} - \Phi_{yx}|_{y-\frac{1}{2}\delta y} = \frac{\partial(\Phi_{yx})}{\partial y}\delta y \quad (83)$$

$$\Phi_{zx}|_{z+\frac{1}{2}\delta z} - \Phi_{zx}|_{z-\frac{1}{2}\delta z} = \frac{\partial(\Phi_{zx})}{\partial z}\delta z \quad (84)$$

These fluxes must be multiplied by the areas of the faces on which they insist before they can be included in the balance equation.

In order to write equations that are as general as possible, we also introduce the possibility of external forces acting on the volume and which give an additional contribution to the momentum flux. For example, the fluid weight is one of these forces, but we can also introduce additional forces. Then, let $\mathbf{F}$ denote this specific external force. Since the fluid weight will never be null, we can divide $\mathbf{F}$ into two contributions, the first of which is associated with fluid weight (through gravity acceleration $\mathbf{g}$):

$$\rho\mathbf{F} = \rho\mathbf{g} + \rho\mathbf{f} \quad (85)$$

This external force induces a momentum flux:

$$\mathbf{F}\delta x\delta y\delta z \quad (86)$$

Now we can construct the momentum balance:

$$\frac{\partial(\rho v_x)}{\partial t}\delta x\delta y\delta z = -\left[\frac{\partial(\Phi_{xx})}{\partial x} + \frac{\partial(\Phi_{yx})}{\partial y} + \frac{\partial(\Phi_{zx})}{\partial z}\right]\delta x\delta y\delta z + \rho F_x\delta x\delta y\delta z \quad (87)$$

$$\frac{\partial(\rho v_x)}{\partial t} = -\left[\frac{\partial(\Phi_{xx})}{\partial x} + \frac{\partial(\Phi_{yx})}{\partial y} + \frac{\partial(\Phi_{zx})}{\partial z}\right] + \rho F_x \quad (88)$$

In order to have a more compact notation, we introduce the divergence:

$$\frac{\partial(\rho v_x)}{\partial t} = -\nabla \cdot \Phi_x^* + \rho F_x \quad (89)$$

where Φ_x^* is the $(\Phi_{xx}, \Phi_{yx}, \Phi_{zx})$ vector. Similarly, for the remaining two directions:

$$\frac{\partial(\rho v_y)}{\partial t} = -\nabla \cdot \Phi_y^* + \rho F_y \quad (90)$$

$$\frac{\partial(\rho v_z)}{\partial t} = -\nabla \cdot \Phi_z^* + \rho F_z \quad (91)$$

Or, in a more compact form:

$$\frac{\partial(\rho v_i)}{\partial t} = -\nabla \cdot \Phi_i^* + \rho F_i \quad (92)$$

We can also adopt the following notation, perfectly equivalent:

$$\frac{\partial(\rho v_i)}{\partial t} = -[\nabla \cdot \underline{\Phi}]_i + \rho F_i \quad (93)$$

where $[\nabla \cdot \underline{\Phi}]_i$ is the i-th component of $\nabla \cdot \underline{\Phi}$, which is the divergence of tensor $\underline{\Phi}$:

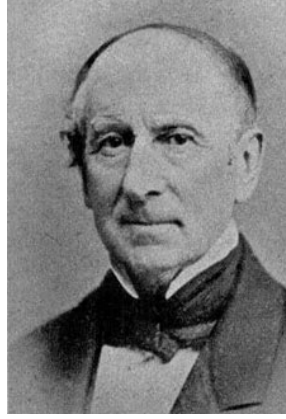


Figure 7: Augustin Louis Cauchy (1789-1857)

$$\nabla \cdot \underline{\Phi} = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) \begin{bmatrix} \Phi_{xx} & \Phi_{xy} & \Phi_{xz} \\ \Phi_{yx} & \Phi_{yy} & \Phi_{yz} \\ \Phi_{zx} & \Phi_{zy} & \Phi_{zz} \end{bmatrix} = \begin{bmatrix} \frac{\partial \Phi_{xx}}{\partial x} + \frac{\partial \Phi_{yx}}{\partial y} + \frac{\partial \Phi_{zx}}{\partial z} \\ \frac{\partial \Phi_{xy}}{\partial x} + \frac{\partial \Phi_{yy}}{\partial y} + \frac{\partial \Phi_{zy}}{\partial z} \\ \frac{\partial \Phi_{xx}}{\partial x} + \frac{\partial \Phi_{yz}}{\partial y} + \frac{\partial \Phi_{zz}}{\partial z} \end{bmatrix} \quad (94)$$

We can write the three scalar equations above as a single vector equation:

$$\frac{\partial (\rho \mathbf{v})}{\partial t} = -\nabla \cdot \underline{\Phi} + \rho \mathbf{F} \quad (95)$$

Usually, it is more convenient to explicitly write the three different tensors:

$$\frac{\partial (\rho \mathbf{v})}{\partial t} = -\nabla p - \nabla \cdot \underline{\tau} - \nabla \cdot (\rho \mathbf{v} \otimes \mathbf{v}) + \rho \mathbf{F} \quad (96)$$

where $\otimes$ is called the outer product, defined as $\mathbf{u} \otimes \mathbf{v} = \mathbf{u} \mathbf{v}^T$. This is the conservation of momentum or *Cauchy's Equation*.

Often, the same equation is written by introducing the substantial or Lagrangian derivative. In particular:

$$\frac{\partial (\rho \mathbf{v})}{\partial t} + \nabla \cdot (\rho \mathbf{v} \otimes \mathbf{v}) = -\nabla p - \nabla \cdot \underline{\tau} + \rho \mathbf{F} \quad (97)$$

$$\rho \frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \frac{\partial \rho}{\partial t} + \mathbf{v} \nabla \cdot (\rho \mathbf{v}) + \rho \mathbf{v} \nabla \cdot \mathbf{v} = \rho \left[\frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \nabla \cdot \mathbf{v} \right] + \mathbf{v} \left[\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) \right] \quad (98)$$

Then, since $\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0$ because of the continuity equation:

$$\rho \frac{D\mathbf{v}}{Dt} = -\nabla p - \nabla \cdot \underline{\tau} + \rho \mathbf{F} \quad (99)$$

Here below, the Cauchy's Equation is written explicitly for convenience in a 3D Cartesian system:

$$\begin{aligned}
& \begin{bmatrix} \frac{\partial(\rho v_x)}{\partial t} \\ \frac{\partial(\rho v_y)}{\partial t} \\ \frac{\partial(\rho v_z)}{\partial t} \end{bmatrix} + \begin{bmatrix} \frac{\partial(\rho v_x v_x)}{\partial x} + \frac{\partial(\rho v_y v_x)}{\partial y} + \frac{\partial(\rho v_z v_x)}{\partial z} \\ \frac{\partial(\rho v_x v_y)}{\partial x} + \frac{\partial(\rho v_y v_y)}{\partial y} + \frac{\partial(\rho v_z v_y)}{\partial z} \\ \frac{\partial(\rho v_x v_z)}{\partial x} + \frac{\partial(\rho v_y v_z)}{\partial y} + \frac{\partial(\rho v_z v_z)}{\partial z} \end{bmatrix} = \\
& = - \begin{bmatrix} \frac{\partial p}{\partial x} \\ \frac{\partial p}{\partial y} \\ \frac{\partial p}{\partial z} \end{bmatrix} - \begin{bmatrix} \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} \\ \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} + \frac{\partial \tau_{zy}}{\partial z} \\ \frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \tau_{zz}}{\partial z} \end{bmatrix} + \rho \begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix}
\end{aligned}$$

Mechanical energy equation From the equation of conservation of momentum, it is possible to obtain the equation of mechanical energy. This is not a real quantity which is conserved, but it is extremely useful in many cases. The equation of mechanical energy is obtained by multiplying by velocity $\mathbf{v}$ the momentum equation (99):

$$\mathbf{v} \cdot \rho \frac{D\mathbf{v}}{Dt} = -\mathbf{v} \cdot \nabla p - \mathbf{v} \cdot (\nabla \cdot \underline{\tau}) + \mathbf{v} \cdot (\rho \mathbf{F}) \quad (100)$$

It is easy to demonstrate that:

$$\mathbf{v} \cdot \rho \frac{D\mathbf{v}}{Dt} = \rho \frac{D}{Dt} \left(\frac{1}{2} \mathbf{v} \cdot \mathbf{v} \right) = \rho \frac{D\hat{E}_K}{Dt} \quad (101)$$

where $\hat{E}_K$ is the specific kinetic energy (see next section). Thus, the equation of mechanical energy is:

$$\rho \frac{D\hat{E}_K}{Dt} = -\mathbf{v} \cdot \nabla p - \mathbf{v} \cdot \nabla \cdot \underline{\tau} + \rho \mathbf{F} \cdot \mathbf{v} \quad (102)$$

6 Conservation of energy

Let us consider all the possible contributions to the definition of total energy of a fluid:

- *kinetic energy*: associated with fluid velocity; In the following we will refer to the specific kinetic energy $\hat{E}_K$ defined as $\hat{E}_K = \frac{1}{2} \mathbf{v} \cdot \mathbf{v}$;
- *internal energy*: associated with the molecular motion and ultimately to the temperature; in particular it will be assumed that the specific internal energy $\hat{U}$ is a function of temperature, density, and fluid composition as if the fluid was in thermodynamic equilibrium $\hat{U} = \hat{U}(\rho, T, \omega_j)$;
- *potential energy*: associated to the gravitational field or more in general to external forces; in general, it is preferred to treat potential energy differently from kinetic and internal ones, and rather to associate it with the work done on the system by the gravitational field and hence as a sort of external contribution.

Thus, we consider the potential energy as an external source term and we focus the attention on kinetic energy and internal energy. We use the Eulerian approach and thus we refer to the cube shown in Figure 8. Take one of the six faces and let us write the flux of energy crossing it (i.e. the flux of convective energy):

$$\mathbf{e}_{conv} = \rho (\hat{U} + \hat{E}_K) \mathbf{v} \quad (103)$$

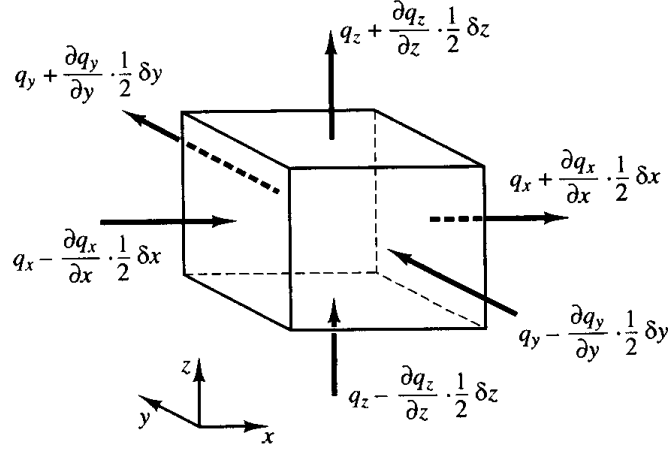


Figure 8: Control volume for writing the energy balance equation

We must also take into account the terms associated with the fluid stresses. Let us consider as usual a reference face perpendicular to the x axis, on which we have the velocity $\mathbf{v}$ and the tensor of molecular fluxes $\underline{\pi}$ (which is a force per unit of surface). In general, the work done by a force $\mathbf{F}$ over a spatial step $d\mathbf{r}$ is expressed as:

$$dL = \mathbf{F} \cdot d\mathbf{r} \quad (104)$$

The corresponding power is:

$$\frac{dL}{dt} = \mathbf{F} \cdot \frac{d\mathbf{r}}{dt} = \mathbf{F} \cdot \mathbf{v} \quad (105)$$

Thus we can associate the following energy flux to the molecular stresses:

$$\mathbf{e}_{stress} = \underline{\pi} \mathbf{v} \quad (106)$$

In order to better clarify the product above, we can write it explicitly in a Cartesian system:

$$\mathbf{e}_{stress} = \underline{\pi} \mathbf{v} = \begin{bmatrix} \pi_x \mathbf{v} \\ \pi_y \mathbf{v} \\ \pi_z \mathbf{v} \end{bmatrix} = \begin{bmatrix} \pi_{xx} v_x + \pi_{xy} v_y + \pi_{xz} v_z \\ \pi_{yx} v_x + \pi_{yy} v_y + \pi_{yz} v_z \\ \pi_{zx} v_x + \pi_{zy} v_y + \pi_{zz} v_z \end{bmatrix} \quad (107)$$

Or, alternatively:

$$\mathbf{e}_{stress} = \underline{\pi} \mathbf{v} = (\pi_x \cdot \mathbf{v}) \delta_x + (\pi_y \cdot \mathbf{v}) \delta_y + (\pi_z \cdot \mathbf{v}) \delta_z \quad (108)$$

Let us move to the diffusion fluxes of energy, similarly to what we did for mass and momentum transfer. Let us call this contribution as $\mathbf{q}$.

Thus, the total flux of energy is:

$$\mathbf{e} = \rho (\hat{U} + \hat{E}_K) \mathbf{v} + \underline{\pi} \mathbf{v} + \mathbf{q} \quad (109)$$

As usual, we can now write the balance over the single volume:

$$\begin{aligned} & \left[\begin{array}{c} \text{accumulation} \\ \text{of energy} \end{array} \right] = \left[\begin{array}{c} \text{net convective inlet} \\ \text{flow rate of energy} \end{array} \right] + \\ & + \left[\begin{array}{c} \text{net diffusion flow} \\ \text{rate of energy} \end{array} \right] + \left[\begin{array}{c} \text{energy flow rate} \\ \text{associated to stresses} \end{array} \right] + \left[\begin{array}{c} \text{external} \\ \text{work} \end{array} \right] \end{aligned}$$

We added in the balance above a source term accounting for the work associated to possible external forces acting on the system. The accumulation term, as usual, is:

$$\frac{\partial \left(\rho \left(\hat{U} + \hat{E}_K \right) \right)}{\partial t} \delta x \delta y \delta z \quad (110)$$

Similarly to what already done for mass and momentum transfer, we can write the inflow and outflow terms. In particular, for faces perpendicular to the x axis:

$$e_x|_{x-\frac{1}{2}\delta x} \simeq e_x - \frac{\partial e_x}{\partial x} \frac{1}{2} \delta x \quad (111)$$

$$e_x|_{x+\frac{1}{2}\delta x} \simeq e_x + \frac{\partial e_x}{\partial x} \frac{1}{2} \delta x \quad (112)$$

The net difference is:

$$e_x|_{x+\frac{1}{2}\delta x} - e_x|_{x-\frac{1}{2}\delta x} = \frac{\partial e_x}{\partial x} \delta x \quad (113)$$

Similarly, for the remaining two directions:

$$e_y|_{y+\frac{1}{2}\delta y} - e_y|_{y-\frac{1}{2}\delta y} = \frac{\partial e_y}{\partial y} \delta y \quad (114)$$

$$e_z|_{z+\frac{1}{2}\delta z} - e_z|_{z-\frac{1}{2}\delta z} = \frac{\partial e_z}{\partial z} \delta z \quad (115)$$

Thus, we have the conservation equation of energy:

$$\frac{\partial \left(\rho \left(\hat{U} + \hat{E}_K \right) \right)}{\partial t} \delta x \delta y \delta z = - \left[\frac{\partial e_x}{\partial x} + \frac{\partial e_y}{\partial y} + \frac{\partial e_z}{\partial z} \right] \delta x \delta y \delta z \quad (116)$$

Here, we neglected for simplicity the additional source term associated to possible external forces. For sure, for real problems, the gravitational field is always present, so we have at least to include $\rho \mathbf{F} \mathbf{v}$. Thus, a more correct version of the equation reported above is:

$$\frac{\partial}{\partial t} \left[\rho \left(\hat{U} + \hat{E}_K \right) \right] = -\nabla \cdot \mathbf{e} + \rho \mathbf{F} \cdot \mathbf{v} \quad (117)$$

This can be also rewritten in explicit form:

$$\frac{\partial}{\partial t} \left[\rho \left(\hat{U} + \hat{E}_K \right) \right] + \nabla \cdot \left[\rho \left(\hat{U} + \hat{E}_K \right) \mathbf{v} \right] = -\nabla \cdot (\underline{\pi} \mathbf{v}) - \nabla \cdot \mathbf{q} + \rho \mathbf{F} \cdot \mathbf{v} \quad (118)$$

Or:

$$\frac{\partial}{\partial t} \left[\rho \left(\hat{U} + \hat{E}_K \right) \right] + \nabla \cdot \left[\rho \left(\hat{U} + \hat{E}_K \right) \mathbf{v} \right] = -\nabla \cdot (p \underline{\mathbf{d}} \mathbf{v}) - \nabla \cdot (\underline{\tau} \mathbf{v}) - \nabla \cdot \mathbf{q} + \rho \mathbf{f} \cdot \mathbf{v} + \rho \mathbf{g} \cdot \mathbf{v} \quad (119)$$

The seven terms that appear in the above equation have a clear and easily identifiable physical meaning: the first identifies the rate of energy accumulation per unit of volume (p.u.v.); the second term is the rate of increase of energy p.u.v due to the convective transport (i.e. the one associated with the fluid dynamic velocity); the second member terms represent the rate of increase of kinetic energy p.u.v. due to pressure forces, viscous forces, external forces other than the gravitational field and the force associated with the gravitational field, respectively.

Observation I The conservation equation of energy can be written also directly using the concept of potential energy. Indeed, if $\hat{\Phi}$ is the potential energy per unit of mass, we know that:

$$\mathbf{g} = -\nabla\hat{\Phi} \quad (120)$$

Thus, the source term in the energy balance equation is:

$$\rho\mathbf{g} \cdot \mathbf{v} = -\rho\mathbf{v} \cdot \nabla\hat{\Phi} = -\nabla \cdot (\rho\mathbf{v}\hat{\Phi}) + \hat{\Phi}\nabla \cdot (\rho\mathbf{v}) = -\nabla \cdot (\rho\mathbf{v}\hat{\Phi}) - \hat{\Phi}\frac{\partial\rho}{\partial t} \quad (121)$$

But $\hat{\Phi}$ is independent of time, thus:

$$\rho\mathbf{g} \cdot \mathbf{v} = -\nabla \cdot (\rho\mathbf{v}\hat{\Phi}) - \frac{\partial(\rho\hat{\Phi})}{\partial t} \quad (122)$$

Therefore, we have the final version of the energy balance equation:

$$\frac{\partial}{\partial t} [\rho(\hat{U} + \hat{E}_K + \hat{\Phi})] + \nabla \cdot [\rho(\hat{U} + \hat{E}_K + \hat{\Phi})\mathbf{v}] = -\nabla \cdot (p\mathbf{\underline{\delta}}\mathbf{v}) - \nabla \cdot (\mathbf{\underline{\tau}}\mathbf{v}) - \nabla \cdot \mathbf{q} + \rho\mathbf{f} \cdot \mathbf{v} \quad (123)$$

Observation II Similarly, the energy for mechanical energy (102) can be rewritten by explicitly using the potential energy:

$$\rho\frac{D}{Dt}(\hat{E}_K + \hat{\Phi}) = -\mathbf{v} \cdot \nabla p - \mathbf{v} \cdot (\nabla \cdot \mathbf{\underline{\tau}}) + \rho\mathbf{f} \cdot \mathbf{v} \quad (124)$$

Case I: pure fluids

The equation written above however is not very convenient to use for practical applications. With some mathematical manipulations, it is possible to get a more usable, equivalent, formulation. Let us start by considering the case of a pure fluid and then we will see what happens if we go to consider the most general case of a mixture.

Let us start from the energy balance equation:

$$\begin{aligned} & \frac{\partial}{\partial t} (\rho(\hat{U} + \hat{E}_K)) + \nabla \cdot (\rho(\hat{U} + \hat{E}_K)\mathbf{v}) = \\ & = \rho\frac{\partial}{\partial t}(\hat{U} + \rho\hat{E}_K) + (\hat{U} + \rho\hat{E}_K)\frac{\partial\rho}{\partial t} + (\hat{U} + \hat{E}_K)\nabla \cdot (\rho\mathbf{v}) + \rho\mathbf{v} \cdot \nabla(\hat{U} + \hat{E}_K) \end{aligned}$$

$$\begin{aligned} & \frac{\partial}{\partial t} (\rho(\hat{U} + \hat{E}_K)) + \nabla \cdot (\rho(\hat{U} + \hat{E}_K)\mathbf{v}) = \\ & = (\hat{U} + \rho\hat{E}_K) \left[\frac{\partial\rho}{\partial t} + \nabla \cdot (\rho\mathbf{v}) \right] + \rho \left[\frac{\partial}{\partial t}(\hat{U} + \rho\hat{E}_K) + \mathbf{v} \cdot \nabla(\hat{U} + \hat{E}_K) \right] = \rho\frac{D}{Dt}(\hat{U} + \hat{E}_K) \end{aligned}$$

Thanks to the continuity equation:

$$\rho \frac{D}{Dt} (\hat{U} + \hat{E}_K) = -\nabla \cdot (\underline{\tau} \mathbf{v}) - \nabla \cdot \mathbf{q} + \rho \mathbf{F} \cdot \mathbf{v} \quad (125)$$

Let us split the molecular stress tensor in its two components:

$$\rho \frac{D}{Dt} (\hat{U} + \hat{E}_K) = -\nabla \cdot (p \underline{\delta} \mathbf{v}) - \nabla \cdot (\underline{\tau} \mathbf{v}) - \nabla \cdot \mathbf{q} + \rho \mathbf{F} \cdot \mathbf{v} \quad (126)$$

Let us subtract from the equation above the equation of mechanical energy (102):

$$\rho \frac{D\hat{U}}{Dt} = -\nabla \cdot (\underline{\tau} \mathbf{v}) - \nabla \cdot (p \mathbf{v}) - \nabla \cdot \mathbf{q} + \rho \mathbf{F} \cdot \mathbf{v} + \mathbf{v} \cdot \nabla p + \mathbf{v} \cdot (\nabla \cdot \underline{\tau}) - \rho \mathbf{F} \cdot \mathbf{v} \quad (127)$$

Thus:

$$\rho \frac{D\hat{U}}{Dt} = -p \cdot \nabla \mathbf{v} - \underline{\tau} : \nabla \mathbf{v} - \nabla \cdot \mathbf{q} \quad (128)$$

To get the equation above, we exploited the two properties of $\nabla \cdot$:

$$\nabla \cdot (\underline{\tau} \mathbf{v}) = \underline{\tau} : \nabla \mathbf{v} + \mathbf{v} \cdot (\nabla \cdot \underline{\tau}) \quad (129)$$

$$\nabla \cdot (p \mathbf{v}) = p \nabla \cdot \mathbf{v} + \mathbf{v} \cdot \nabla p \quad (130)$$

The term $-\underline{\tau} : \nabla \mathbf{v}$ is always a positive quantity (it can be demonstrated analytically) and represents the heat dissipated by the viscous forces Q_{diss} . In Cartesian coordinates, it is given by:

$$Q_{diss} = -\underline{\tau} : \nabla \mathbf{v} = -\tau_{xx} \frac{\partial v_x}{\partial x} - \tau_{xy} \frac{\partial v_y}{\partial x} - \tau_{xz} \frac{\partial v_z}{\partial x} - \tau_{yx} \frac{\partial v_x}{\partial y} + \\ -\tau_{yy} \frac{\partial v_y}{\partial y} - \tau_{yz} \frac{\partial v_z}{\partial y} - \tau_{zx} \frac{\partial v_x}{\partial z} - \tau_{zy} \frac{\partial v_y}{\partial z} - \tau_{zz} \frac{\partial v_z}{\partial z}$$

This term represents the degradation of mechanical energy in thermal energy that occurs in all moving fluids. This term can lead to significant temperature increases only if the viscosity is very high or at extremely high speed gradients.

Let's now introduce the enthalpy function, defined as follows:

$$\hat{H} = \hat{U} + \frac{p}{\rho} \quad (131)$$

Its substantial derivative is:

$$\frac{D\hat{H}}{Dt} = \frac{D\hat{U}}{Dt} + \frac{1}{\rho} \frac{Dp}{Dt} - \frac{p}{\rho^2} \frac{D\rho}{Dt} \quad (132)$$

$$\rho \frac{D\hat{H}}{Dt} = \rho \frac{D\hat{U}}{Dt} + \frac{Dp}{Dt} - \frac{p}{\rho} \frac{D\rho}{Dt} = \rho \frac{D\hat{U}}{Dt} + \frac{Dp}{Dt} + p \nabla \cdot \mathbf{v} \quad (133)$$

If we put the expression above in the energy balance equation, we have:

$$\rho \frac{D\hat{H}}{Dt} = \frac{Dp}{Dt} + Q_{diss} - \nabla \cdot \mathbf{q} \quad (134)$$

For pure fluids, enthalpy is a function of temperature and pressure only. This means that:

$$\rho \frac{D\hat{H}}{Dt} = \rho \left[\frac{D\hat{H}}{DT} \frac{DT}{Dt} + \frac{D\hat{H}}{Dp} \frac{Dp}{Dt} \right] = \rho \left[C_P \frac{DT}{Dt} + \frac{1 - \beta T}{\rho} \frac{Dp}{Dt} \right] = \rho C_P \frac{DT}{Dt} + (1 - \beta T) \frac{Dp}{Dt} \quad (135)$$

where β is the compressibility factor[†]:

$$\beta = -\frac{1}{\rho} \frac{\partial \rho}{\partial T} \quad (136)$$

Thus, we can have the same energy equation, but written in terms of temperature instead of enthalpy:

$$\rho C_P \frac{DT}{Dt} = \beta T \frac{Dp}{Dt} + Q_{diss} - \nabla \cdot \mathbf{q} \quad (137)$$

Case II: multicomponent fluids

Let us therefore consider the most general case of a multi-component fluid. The main difference is that internal energy is obviously a function of composition (unlike kinetic energy, which depends on speed only):

$$\rho \hat{U} = \sum_j \rho_j \hat{U}_j \quad (138)$$

It is more convenient to rewrite the convective flux in the following, equivalent form, more useful:

$$\mathbf{e}_{conv} = \rho \hat{E}_K \mathbf{v} + \sum_j \rho_j \mathbf{v}_j \hat{U}_j \quad (139)$$

Thus, the total flux of energy is:

$$\mathbf{e} = \rho \hat{E}_K \mathbf{v} + \sum_j \rho_j \mathbf{v}_j \hat{U}_j + \nabla \cdot (\underline{\pi} \mathbf{v}) - \nabla \cdot \mathbf{q} \quad (140)$$

The $\rho \mathbf{F} \mathbf{v}$ can be also rewritten as:

$$\sum_j \rho_j \mathbf{F}_j \cdot \mathbf{v}_j = \sum_j \mathbf{F}_j \cdot (\rho_j \mathbf{v} + \mathbf{j}_j) = \sum_j \rho_j \mathbf{F}_j \cdot \mathbf{v} + \sum_j \mathbf{F}_j \cdot \mathbf{j}_j = \rho \mathbf{F} \cdot \mathbf{v} + \sum_j \mathbf{F}_j \cdot \mathbf{j}_j \quad (141)$$

Thus, the energy balance equation becomes:

$$\frac{\partial}{\partial t} \left(\rho (\hat{U} + \hat{E}_K) \right) = -\nabla \cdot (\rho \hat{E}_K \mathbf{v}) - \nabla \cdot \left(\sum_j \rho_j \mathbf{v}_j \hat{U}_j \right) - \nabla \cdot (\underline{\pi} \mathbf{v}) - \nabla \cdot \mathbf{q} + \rho \mathbf{F} \cdot \mathbf{v} + \sum_j \mathbf{F}_j \cdot \mathbf{j}_j \quad (142)$$

Let us focus the attention of the second contribution on the RHS:

$$\sum_j \rho_j \mathbf{v}_j \hat{U}_j = \sum_j \hat{U}_j (\rho_j \mathbf{v} + \mathbf{j}_j) = \sum_j \hat{U}_j \rho_j \mathbf{v} + \sum_j \hat{U}_j \mathbf{j}_j = \rho \hat{U} \mathbf{v} + \sum_j \hat{U}_j \mathbf{j}_j \quad (143)$$

This allows us to rewrite the energy balance equation as:

$$\frac{\partial}{\partial t} \left(\rho \left(\hat{U} + \hat{E}_K \right) \right) + \nabla \cdot \left(\rho \left(\hat{U} + \hat{E}_K \right) \mathbf{v} \right) = -\nabla \cdot \left(\sum_j \hat{U}_j \mathbf{j}_j \right) - \nabla \cdot (\boldsymbol{\pi} \mathbf{v}) - \nabla \cdot \mathbf{q} + \rho \mathbf{F} \cdot \mathbf{v} + \sum_j \mathbf{F}_j \cdot \mathbf{j}_j \quad (144)$$

The LHS can be rewritten in a more compact form by introducing the substantial derivative and the continuity equation:

$$\rho \frac{D}{Dt} \left(\hat{U} + \hat{E}_K \right) = -\nabla \cdot \left(\sum_j \hat{U}_j \mathbf{j}_j \right) - \nabla \cdot (\boldsymbol{\pi} \mathbf{v}) - \nabla \cdot \mathbf{q} + \rho \mathbf{F} \cdot \mathbf{v} + \sum_j \mathbf{F}_j \cdot \mathbf{j}_j \quad (145)$$

Let us subtract the equation of mechanical energy (102):

$$\rho \frac{D\hat{U}}{Dt} = -\nabla \cdot \left(\sum_j \hat{U}_j \mathbf{j}_j \right) - p \nabla \cdot \mathbf{v} - \boldsymbol{\tau} : \nabla \mathbf{v} - \nabla \cdot \mathbf{q} + \sum_j \mathbf{F}_j \cdot \mathbf{j}_j \quad (146)$$

Similarly to what we did for the pure fluid, let us introduce the definition of enthalpy:

$$\rho \frac{D\hat{H}}{Dt} = \frac{Dp}{Dt} - \nabla \cdot \left(\sum_j \hat{U}_j \mathbf{j}_j \right) + Q_{diss} - \nabla \cdot \mathbf{q} + \sum_j \mathbf{F}_j \cdot \mathbf{j}_j \quad (147)$$

Now, since we have a multicomponent fluid, the enthalpy is not only function of temperature and pressure, but also of composition:

$$\rho \frac{D\hat{H}}{Dt} = \rho \left[\frac{D\hat{H}}{DT} \frac{DT}{Dt} + \frac{D\hat{H}}{Dp} \frac{Dp}{Dt} + \sum_j \frac{D\hat{H}}{D\omega_j} \frac{D\omega_j}{Dt} \right] = \rho C_P \frac{DT}{Dt} + (1 - \beta T) \frac{Dp}{Dt} + \rho \sum_j \hat{H}_j \frac{D\omega_j}{Dt} \quad (148)$$

We already derived the conservation equation for species j :

$$\rho \frac{D\omega_j}{Dt} = -\nabla \cdot \mathbf{j}_j + R_j \quad (149)$$

$$\rho \frac{D\hat{H}}{Dt} = \rho C_P \frac{DT}{Dt} + (1 - \beta T) \frac{Dp}{Dt} + \sum_j \hat{H}_j R_j - \sum_j \hat{H}_j \nabla \cdot \mathbf{j}_j \quad (150)$$

After some math:

$$\rho C_P \frac{DT}{Dt} = \beta T \frac{Dp}{Dt} - \nabla \cdot \left(\sum_j \hat{U}_j \mathbf{j}_j \right) + Q_{diss} - \nabla \cdot \mathbf{q} + \sum_j \mathbf{F}_j \cdot \mathbf{j}_j - \sum_j \hat{H}_j R_j + \sum_j \hat{H}_j \nabla \cdot \mathbf{j}_j \quad (151)$$

Usually, the equation above is written by introducing the reaction heat:

$$Q_R = - \sum_j \hat{H}_j R_j \quad (152)$$

On the contrary, the second term on the RHS is rewritten as:

$$\nabla \cdot \left(\sum_j \hat{U}_j \mathbf{j}_j \right) = \nabla \cdot \left(\sum_j \left(\hat{H}_j - \frac{p}{\rho} \right) \mathbf{j}_j \right) = \nabla \cdot \left(\sum_j \hat{H}_j \mathbf{j}_j - \frac{p}{\rho} \sum_j \mathbf{j}_j \right) = \nabla \cdot \left(\sum_j \hat{H}_j \mathbf{j}_j \right) \quad (153)$$

We can sum this term to the last one in the nergy balance equation:

$$\nabla \cdot \left(\sum_j \hat{U}_j \mathbf{j}_j \right) + \sum_j \hat{H}_j \nabla \cdot \mathbf{j}_j = \nabla \cdot \left(\sum_j \hat{H}_j \mathbf{j}_j \right) + \sum_j \hat{H}_j \nabla \cdot \mathbf{j}_j = - \sum_j \mathbf{j}_j \cdot \nabla \hat{H}_j \quad (154)$$

This is the final version of energy equation, written in terms of temperature:

$$\rho C_P \frac{DT}{Dt} = \beta T \frac{Dp}{Dt} - \sum_j \mathbf{j}_j \cdot \nabla \hat{H}_j + Q_{diss} - \nabla \cdot \mathbf{q} + \sum_j \mathbf{F}_j \cdot \mathbf{j}_j + Q_R \quad (155)$$

Observation

In many CFD codes the energy equation is used in the enthalpy formulation. However, not always the formulation we reported above (147) is really adopted. First of all in equation (147) we exploit equation (153), to get the following equation:

$$\rho \frac{D\hat{H}}{Dt} = \frac{Dp}{Dt} - \nabla \cdot \left(\sum_j \hat{H}_j \mathbf{j}_j \right) + Q_{diss} - \nabla \cdot \mathbf{q} + \sum_j \mathbf{F}_j \cdot \mathbf{j}_j \quad (156)$$

Instead of using the total enthalpy $\hat{H}_j$, we split it in its two contributions: the sensible enthalpy $\hat{H}_{j,s}$ and the formation enthalpy $\hat{H}_{j,f}^0$:

$$\hat{H}_{j,s} = \hat{H}_j - \hat{H}_{j,f}^0 \quad (157)$$

$$\hat{H}_s = \hat{H} - \sum_j \hat{H}_{j,f}^0 \quad (158)$$

Thus, the energy balance equation becomes:

$$\rho \frac{D\hat{H}_s}{Dt} = Q_R^0 + \frac{Dp}{Dt} - \nabla \cdot \left(\sum_j \hat{H}_{j,s} \mathbf{j}_j \right) + Q_{diss} - \nabla \cdot \mathbf{q} + \sum_j \mathbf{F}_j \cdot \mathbf{j}_j \quad (159)$$

where Q_R^0 is the heat released by the reactions, which is not exactly Q_R , since it is defined only on the formation enthalpies:

$$Q_R^0 = - \sum_j \hat{H}_{j,f}^0 R_j \quad (160)$$

Equation (159) is the most common version of energy equation adopted in the CFD codes.

7 Constitutive laws

Now we want to identify constitutive laws that quantify the diffusion fluxes of species, energy, and momentum that have been introduced into the corresponding balance equations. In the conservation equations we saw that it is necessary to express such fluxes as a function of variables that characterize the fluid, such as temperature, pressure, density and composition, as well as the physical properties. We will refer hereafter to fluids without memory, for which instantaneous and local state can always be defined in terms of instantaneous and local variables. We will see, that the diffusive fluxes of the three main quantities of interest (species, energy, and momentum) have very similar expressions, being related to the second principle of thermodynamics. Be aware that, unlike the conservation equations, the constitutive laws we are going to define are the mathematical translation of empirical observations and therefore they represent models. Thus they can be more or less detailed. In fact, we will see that constitutive laws can have more terms depending on the level of detail that you want to reach and depending on the type of problem you are considering.

7.1 Mass diffusion fluxes

Let us start with the mass diffusion flux $\mathbf{j}_j$ appearing in the conservation equation of species:

$$\rho \frac{D\omega_j}{Dt} = -\nabla \cdot \mathbf{j}_j + R_j \quad (161)$$

It is better to rewrite the equation above by using the concept of diffusion velocity $\mathbf{v}_{j,d}$:

$$\rho \frac{D\omega_j}{Dt} = \rho \frac{\partial \omega_j}{\partial t} + \rho \mathbf{v} \cdot \nabla \omega_j = \left[\frac{\partial \rho_j}{\partial t} + \nabla \cdot (\rho_j \mathbf{v}) \right] - \omega_j \left[\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) \right] = \frac{\partial \rho_j}{\partial t} + \nabla \cdot (\rho_j \mathbf{v}) \quad (162)$$

$$\frac{\partial \rho_j}{\partial t} + \nabla \cdot (\rho_j \mathbf{v} + \mathbf{j}_j) = R_j \quad (163)$$

$$\frac{\partial \rho_j}{\partial t} + \nabla \cdot (\rho \omega_j \mathbf{v} + \rho \omega_j \mathbf{v}_{j,d}) = \frac{\partial \rho_j}{\partial t} + \nabla \cdot (\rho \omega_j (\mathbf{v} + \mathbf{v}_{j,d})) = R_j \quad (164)$$

Thus the conservation equation of species can be written equivalently as:

$$\frac{\partial \rho_j}{\partial t} + \nabla \cdot (\rho \omega_j (\mathbf{v} + \mathbf{v}_{j,d})) = R_j \quad (165)$$

where:

$$\mathbf{j}_j = \rho \omega_j \mathbf{v}_{j,d} \quad (166)$$

The mass diffusion flux is the sum of four main contributions:

- molecular diffusion $\mathbf{j}_j^o$: due to the composition gradients;
- thermal diffusion $\mathbf{j}_j^t$: due to the temperature gradients (*Soret effect*);
- pressure diffusion $\mathbf{j}_j^p$: due to gradients of pressure;
- external diffusion $\mathbf{j}_j^f$: due to external forces acting on the species in a preferential way.

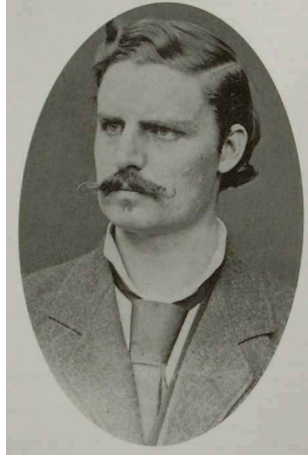


Figure 9: Adolf Eugen Fick (1829-1901)

At first approximation, each of these contributions can be considered proportional to the force (i.e. the gradient) that causes it. Overall, the total mass diffusion flux is therefore:

$$\mathbf{j}_j = \mathbf{j}_j^o + \mathbf{j}_j^t + \mathbf{j}_j^p + \mathbf{j}_j^f \quad (167)$$

Of all these contributions, the most significant one is the ordinary diffusion (at least if we exclude ionic species in the electric and magnetic fields for which forced diffusion is extremely significant and the strong gradients of temperature used in the isotope separations). In the first approximation, as noted above, it is possible to assume that the diffusive flux is simply proportional to the gradient of composition. This is how the famous *Fick's law* is obtained:

$$\mathbf{j}_j = -\rho \Gamma_{j,mix} \nabla \omega_j \quad (168)$$

In the following, we will consider only the molecular contribution, i.e. only the Fick's law. In particular, the Fick's law, in molar terms is:

$$\mathbf{J}_j = -\Gamma_{j,mix} \nabla C_j \quad (169)$$

$\Gamma_{j,mix}$ is the mass diffusion coefficient or diffusivity of species j with respect to the mixture. The Fick's law can be also written by using the diffusion velocity:

$$\omega_j \mathbf{v}_{j,d} = -\Gamma_{j,mix} \nabla \omega_j \quad (170)$$

The diffusivity coefficients are calculated using empirical laws and using appropriate mixing rules. The use of Fick's law has a very significant limit, which can lead to numeric problems if numerical solutions to the transport equations are to be solved. In fact, the sum of diffusion fluxes must be identically equal to zero. So if you use Fick's law it should be:

$$\sum_j^{NC} \mathbf{j}_j = -\rho \sum_j^{NC} \Gamma_{j,mix} \nabla \omega_j = 0 \quad (171)$$

Unfortunately, this closure can be ensured only if $\Gamma_{j,mix}$ are all the same, which is generally not verified. The problem is that the non-zero closure of diffusive flows creates a mass source term in the continuity equation, which is physically unacceptable. There are two possible solutions to this problem:

1. Correction on inert species

We impose the closure of diffusion fluxes by artificially modifying the flux of the most abundant species (typically the inert species, which we imagine the $NC - th$ species) to minimize the error. Thus the equations for the $NC - 1$ species are written as usual, but the equation for the most abundant species is corrected as:

$$\rho \frac{D\omega_{NC}}{Dt} = -\nabla \cdot \mathbf{j}_{NC} + R_{NC} \quad (172)$$

where:

$$\rho \frac{D\omega_{NC}}{Dt} = -\nabla \cdot (\rho \Gamma_{NC,mix} \nabla \omega_{NC}) + R_{NC} \quad (173)$$

$$\rho \Gamma_{NC,mix} \nabla \omega_{NC} + \rho \sum_{j=1}^{NC-1} \Gamma_{j,mix} \nabla \omega_j \neq 0 \quad (174)$$

This approach can be considered acceptable, if the most abundant species chosen for the closure is always present in large amounts (sufficiently larger than zero). However, if we carefully look at the equation of this species, we can recognize the presence of an unphysical term:

$$\begin{aligned} \rho \frac{D\omega_{NC}}{Dt} &= -\nabla \cdot \left(\rho \sum_{j=1}^{NC-1} \Gamma_{j,mix} \nabla \omega_j \right) + R_{NC} = \\ &= -\nabla \cdot \left(\rho \sum_{j=1}^{NC-1} \Gamma_{j,mix} \nabla \omega_j \right) - \nabla \cdot (\rho \Gamma_{NC,mix} \nabla \omega_{NC}) + \nabla \cdot (\rho \Gamma_{NC,mix} \nabla \omega_{NC}) + R_{NC} \\ \rho \frac{D\omega_{NC}}{Dt} &= -\nabla \cdot \left[\rho \left(\sum_{j=1}^{NC-1} \Gamma_{j,mix} \nabla \omega_j + \Gamma_{NC,mix} \nabla \left(1 - \sum_{j=1}^{NC-1} \omega_j \right) \right) \right] + \\ &\quad + \nabla \cdot (\rho \Gamma_{NC,mix} \nabla \omega_{NC}) + R_{NC} \\ \rho \frac{D\omega_{NC}}{Dt} &= -\nabla \cdot \left[\rho \left(\sum_{j=1}^{NC-1} (\Gamma_{j,mix} - \Gamma_{NC,mix}) \nabla \omega_j \right) \right] + \nabla \cdot (\rho \Gamma_{NC,mix} \nabla \omega_{NC}) + R_{NC} \end{aligned} \quad (175)$$

The first contribution on the RHS is the correction, which has no physical meaning.

2. Correction diffusion velocity

The second approach, more physically sound, introduces the so called *correction diffusion velocity* $\mathbf{v}_C$. The idea is to add to the fluid dynamic velocity $\mathbf{v}$ a correction velocity $\mathbf{v}_C$ (usually very small), chosen to ensure the closure of diffusion fluxes. Let us start from the conservation equation of a single species:

$$\frac{\partial \rho_j}{\partial t} + \nabla \cdot (\rho_j \mathbf{v}) = \nabla \cdot (\rho \Gamma_{j,mix} \nabla \omega_j) + R_j \quad (176)$$

Let us add the correction diffusion velocity:

$$\frac{\partial \rho_j}{\partial t} + \nabla \cdot (\rho_j (\mathbf{v} + \mathbf{v}_C)) = \nabla \cdot (\rho \Gamma_{j,mix} \nabla \omega_j) + R_j \quad (177)$$

Its value is obtained by forcing the closure to zero:

$$\sum_j \frac{\partial \rho_j}{\partial t} + \sum_j \nabla \cdot (\rho_j (\mathbf{v} + \mathbf{v}_C)) = \sum_j \nabla \cdot (\rho \Gamma_{j,mix} \nabla \omega_j) + \sum_j R_j \quad (178)$$

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = \nabla \cdot \left(\rho \sum_j (\Gamma_{j,mix} \nabla \omega_j) - \rho \mathbf{v}_C \right) = 0 \quad (179)$$

$$\mathbf{v}_C = \sum_j (\Gamma_{j,mix} \nabla \omega_j) \quad (180)$$

Maxwell-Stefan Equations

Fick's law is the simplest model for expressing the diffusion mass fluxes due to the gradients of composition. Actually, the mass diffusion fluxes should be a function of the gradients of composition of all species, and not just of its own. We should therefore expect a more general law, like the one reported below:

$$\underline{\mathbf{J}} = \begin{bmatrix} \mathbf{J}_1 \\ \mathbf{J}_2 \\ \dots \\ \mathbf{J}_{NC} \end{bmatrix} = -C_{tot} \begin{bmatrix} \Gamma_{11} & \Gamma_{12} & \Gamma_{13} & \dots & \dots \\ \dots & \Gamma_{22} & \dots & \dots & \dots \\ \dots & \dots & \Gamma_{33} & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \Gamma_{NCNC} \end{bmatrix} \begin{bmatrix} \nabla \omega_1 \\ \nabla \omega_2 \\ \dots \\ \dots \\ \nabla \omega_{NC} \end{bmatrix} \quad (181)$$

In a more compact notation:

$$\underline{\mathbf{J}} = -C_{tot} \underline{\mathbf{\Gamma}} \underline{\nabla \omega} \quad (182)$$

While the Fick's law implies that the gradient vector of composition of j and the diffusive flux are always parallel (i.e. aligned), generally a model like the one proposed above does not imply this. Indeed, there are clear examples that the diffusive flux of a species is not always aligned with the gradient of its composition. In some cases, it is also possible to observe a diffusive flow in the opposite direction to that provided by the Fick's law (reverse diffusion). Moreover, Fick's law also implies that a diffusive flux of j exist only if there is a gradient of composition, but experimentally it has been observed that this is not a strictly necessary condition.

Thus, a more accurate model for describing the ordinary diffusion is represented by the so-called *Maxwell-Stefan equations* (the first identified them for a binary system and the second had the merit of extending them to a multi-component system). These equations allow to obtain the diffusive velocities of the different species, but unfortunately they can not be written in an explicit form:

$$\nabla x_k = \sum_j \frac{x_k x_j}{\Gamma_{kj}} (\mathbf{v}_{j,d} - \mathbf{v}_{k,d}) + (\omega_k - X_k) \frac{\nabla p}{p} + \frac{\rho}{p} \sum_j \omega_k \omega_j (\mathbf{F}_k - \mathbf{F}_j) \quad (183)$$



Figure 10: James Clerk Maxwell (1831-1879) and Josef Stefan (1835-1893)

The main disadvantage is therefore that diffusion velocities require the resolution of a system of $NC \times NC$ linear equations. In most CFD cases this additional cost is unacceptable.

Keep in mind that the need to describe very accurately the diffusion fluxes exists only for laminar conditions. In the case of turbulent conditions (as we will better discuss) it is not strictly needed to calculate in a very accurate way the diffusion fluxes of species, which are often neglected.

7.2 Conductive heat flux

Let us now consider the diffusion of energy. The first thing to point out is that we can distinguish between a conductive $\mathbf{q}^{cond}$ and radiative $\mathbf{q}^{rad}$ contribution.

Let us focus the attention on the first one. The conductive contribution is the sum of four sub-contributions:

- $\mathbf{q}^t$: due to temperature gradients;
- $\mathbf{q}^x$: due to composition gradients;
- $\mathbf{q}^p$: due to pressure gradients;
- $\mathbf{q}^f$: due to external forces.

At first approximation, as observed for the diffusion mass fluxes, each of these contributions can be considered proportional to the force that causes it. Overall, the total conductive flux of energy is:

$$\mathbf{q}^{cond} = \mathbf{q}^t + \mathbf{q}^x + \mathbf{q}^p + \mathbf{q}^f \quad (184)$$

In most cases, the only important contribution is the first one, which is given by the *Fourier's law*:

$$\mathbf{q}^t = -k\nabla T \quad (185)$$

where k is the thermal conductivity of the fluid, which depends on temperature, pressure, and composition.

The radiative contribution is much more complex to describe. Its discussion is not elementary as that of conductive contribution and in CFD codes it is usually written



Figure 11: Jean Baptiste Fourier (1768-1830)

using strong simplifications. First of all, we have to consider that gaseous/liquid species and particulate matter in a fluid are generally capable of absorbing and emitting energy, with properties that are of a spectral nature, i.e function of the frequency (or wavelength) of the radiation itself. This is a first element of complexity. Additionally, a portion of the absorbed energy can excite the molecules at different energy levels depending on the characteristics of the incident radiation. All these phenomena become particularly important as the frequency of radiation is higher, so that they can break chemical bonds, when temperatures reach extremely high levels. Even when convection transport is not important, heat transfer by radiation can be particularly significant. In any case, the heat transfer by radiation is capable of influencing the temperature field and more generally the energy distribution only if the medium is opaque, i.e. it is capable of absorbing and emitting in the field of wavelengths relevant for radiation. In addition we have to consider also complications about geometric aspects: in fact the radiative contribution in a point is not only a function of the physical properties of the fluid and the gradients (temperature, for example) at that point, but also a direct function of the geometry of the system (think of the presence of high temperature walls). The problem is therefore extremely complex to be treated in complete form. It should be expressed in the form of integro-differential equations that are solved by extremely sophisticated numeric techniques. If you want to simplify the problem, you can consider this basic rule: if there are no solid bodies/surfaces whose surface is at a high temperature (which can then exchange large amounts of heat by radiation), it is possible to consider only the emission of heat by radiation from a gas according to the following very approximate expression:

$$\nabla \cdot \mathbf{q}^{rad} = a_p \sigma T^4 \quad (186)$$

In the equation above σ is the constant of Stefan-Boltzmann, T is the temperature at which the fluid is located and a_p is the so-called gas or gas absorption coefficient. This is a function of temperature, pressure and composition, which is generally described as the sum of individual contributions associated with each species constituting the system according to a mixing rule based on molar fractions or partial pressures, as follows:

$$a_p = \sum_j a_{p,j} p_j \quad (187)$$

It should be taken into account that in general the gaseous species that play the most significant role in the heat exchange problems by radiation are: H_2O , CO_2 , CO , O_2 and soot or carbonaceous particulate.

7.3 Momentum diffusion fluxes (viscous tensor)

We now focus our attention on the viscous stress tensor, which exists only when the fluid is moving, unlike the hydrostatic tensor (associated with pressure) that exists even when the fluid is at rest. When part of a fluid moves, intermolecular forces that exist due to friction phenomena tend to drag adjacent fluid molecules in the same direction. The magnitude of these intermolecular forces depends on the nature of the fluid and determines the dissipation rate of the internal energy.

We will focus the attention only on Newtonian fluids, i.e. fluids whose viscous stress tensor is given by the *Newton's law*. The Newton's law is the result of a complex analysis that leads to the following law:

$$\underline{\tau} = -\mu \left(\nabla \mathbf{v} + (\nabla \mathbf{v})^T \right) + \left(\frac{2}{3}\mu - \kappa \right) (\nabla \cdot \mathbf{v}) \underline{\delta} \quad (188)$$

where $\underline{\delta}$ is the unit tensor, $\nabla \mathbf{v}$ is the velocity gradient tensor and $(\nabla \mathbf{v})^T$ is its transpose and $\nabla \cdot \mathbf{v}$ simply the divergence of velocity.

In the equation above $\nabla \mathbf{v}$ is:

$$\nabla \mathbf{v} = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right)^T (v_x, v_y, v_z) = \begin{bmatrix} \frac{\partial v_x}{\partial x} & \frac{\partial v_y}{\partial x} & \frac{\partial v_z}{\partial x} \\ \frac{\partial v_x}{\partial y} & \frac{\partial v_y}{\partial y} & \frac{\partial v_z}{\partial y} \\ \frac{\partial v_x}{\partial z} & \frac{\partial v_y}{\partial z} & \frac{\partial v_z}{\partial z} \end{bmatrix} \quad (189)$$

The following quantity, appearing in the Newton's law (188), is called tensor of deformation velocities $\underline{D}$:

$$\underline{D} = \left(\nabla \mathbf{v} + (\nabla \mathbf{v})^T \right) = \begin{bmatrix} 2\frac{\partial v_x}{\partial x} & \frac{\partial v_y}{\partial x} + \frac{\partial v_x}{\partial y} & \frac{\partial v_z}{\partial x} + \frac{\partial v_x}{\partial z} \\ \frac{\partial v_x}{\partial y} + \frac{\partial v_y}{\partial x} & 2\frac{\partial v_y}{\partial y} & \frac{\partial v_z}{\partial y} + \frac{\partial v_y}{\partial z} \\ \frac{\partial v_x}{\partial z} + \frac{\partial v_z}{\partial x} & \frac{\partial v_y}{\partial z} + \frac{\partial v_z}{\partial y} & 2\frac{\partial v_z}{\partial z} \end{bmatrix} \quad (190)$$

$$D_{ij} = \frac{\partial v_j}{\partial x_i} + \frac{\partial v_i}{\partial x_j} \quad (191)$$

Note that a slightly different definition of $\underline{D}$ is often found in the literature (often with a $\frac{1}{2}$ factor).

We see that in the Newton's law two phenomenological coefficients appear: the dynamic viscosity μ and the dilatation viscosity κ . In general, however, it is not necessary to know κ , because for gases it can be assumed to be equal to zero (actually this result is strictly valid only if the gas is monoatomic but in general the error is negligible for polyatomic gases). In the case of liquid fluids, since in most cases we assume the fluid is incompressible, i.e. $\nabla \cdot \mathbf{v} = 0$, the term in which the dilatation viscosity appears is null. The dilatation viscosity is important only in the propagation of pressure waves in the polyatomic gases or in liquid-gas multi-phase systems. The Newton's law in explicit terms is:

$$\tau_{ij} = -\mu \left(\frac{\partial v_j}{\partial x_i} + \frac{\partial v_i}{\partial x_j} \right) + \left(\frac{2}{3}\mu - \kappa \right) \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right) \delta_{ij} \quad (192)$$



Figure 12: Claude Louis Navier (1785-1836) and George Stokes (1819-1903)

$$\begin{bmatrix} \tau_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \tau_{yy} & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \tau_{zz} \end{bmatrix} = -\mu \begin{bmatrix} 2\frac{\partial v_x}{\partial x} & \frac{\partial v_y}{\partial x} + \frac{\partial v_x}{\partial y} & \frac{\partial v_z}{\partial x} + \frac{\partial v_x}{\partial z} \\ \frac{\partial v_x}{\partial y} + \frac{\partial v_y}{\partial x} & 2\frac{\partial v_y}{\partial y} & \frac{\partial v_z}{\partial y} + \frac{\partial v_y}{\partial z} \\ \frac{\partial v_x}{\partial z} + \frac{\partial v_z}{\partial x} & \frac{\partial v_y}{\partial z} + \frac{\partial v_z}{\partial y} & 2\frac{\partial v_z}{\partial z} \end{bmatrix} + \\ + \left(\frac{2}{3}\mu - \kappa\right) \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}\right) \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Navier-Stokes Equations

If the Newton's law (188) is used within the Cauchy's equation (99), we obtain a set of three equations describing the conservation of momentum. If we add also the continuity equation, we have the *Navier-Stokes equations*.

We focus on the x component only; obviously the other two components can be treated in the same way. As we have seen, the equation is as follows:

$$\rho \frac{Dv_x}{Dt} = -\frac{\partial p}{\partial x} - \nabla \cdot \boldsymbol{\tau}_x + \rho F_x = -\frac{\partial p}{\partial x} - \frac{\partial \tau_{xx}}{\partial x} - \frac{\partial \tau_{yx}}{\partial y} - \frac{\partial \tau_{zx}}{\partial z} + \rho F_x \quad (193)$$

$$\begin{aligned} \rho \frac{Dv_x}{Dt} = & -\frac{\partial p}{\partial x} + \frac{\partial}{\partial x} \left[2\mu \frac{\partial v_x}{\partial x} - \left(\frac{2}{3}\mu - \kappa\right) (\nabla \cdot \mathbf{v}) \right] + \\ & + \frac{\partial}{\partial y} \left[\mu \left(\frac{\partial v_y}{\partial x} + \frac{\partial v_x}{\partial y} \right) \right] + \frac{\partial}{\partial z} \left[\mu \left(\frac{\partial v_z}{\partial x} + \frac{\partial v_x}{\partial z} \right) \right] + \rho F_x \end{aligned}$$

Viscous terms are usually grouped together according to the following:

$$\begin{aligned} \rho \frac{Dv_x}{Dt} = & -\frac{\partial p}{\partial x} + \left\{ \frac{\partial}{\partial x} \left(\mu \frac{\partial v_x}{\partial x} \right) + \frac{\partial}{\partial y} \left(\mu \frac{\partial v_x}{\partial y} \right) + \frac{\partial}{\partial z} \left(\mu \frac{\partial v_x}{\partial z} \right) \right\} + \\ & + \left\{ \frac{\partial}{\partial x} \left(\mu \frac{\partial v_x}{\partial x} \right) + \frac{\partial}{\partial y} \left(\mu \frac{\partial v_y}{\partial y} \right) + \frac{\partial}{\partial z} \left(\mu \frac{\partial v_z}{\partial z} \right) \right\} - \frac{\partial}{\partial x} \left[\left(\frac{2}{3}\mu - \kappa\right) (\nabla \cdot \mathbf{v}) \right] + \rho F_x \end{aligned}$$

$$\begin{aligned} \rho \frac{Dv_x}{Dt} = & -\frac{\partial p}{\partial x} + \nabla \cdot (\nabla v_x) + \left\{ \frac{\partial}{\partial x} \left(\mu \frac{\partial v_x}{\partial x} \right) + \frac{\partial}{\partial y} \left(\mu \frac{\partial v_y}{\partial x} \right) + \frac{\partial}{\partial z} \left(\mu \frac{\partial v_z}{\partial x} \right) \right\} - \\ & - \frac{\partial}{\partial x} \left[\left(\frac{2}{3}\mu - \kappa\right) (\nabla \cdot \mathbf{v}) \right] + \rho F_x \end{aligned}$$

$$\rho \frac{Dv_x}{Dt} = -\frac{\partial p}{\partial x} + \nabla \cdot (\mu \nabla v_x) + S_x + \rho F_x \quad (194)$$

Terms collected in S_x are usually written separately, because there are specific numerical algorithms devoted to them during the numerical solution of Navier-Stokes equations:

$$S_x = \left\{ \frac{\partial}{\partial x} \left(\mu \frac{\partial v_x}{\partial x} \right) + \frac{\partial}{\partial y} \left(\mu \frac{\partial v_y}{\partial x} \right) + \frac{\partial}{\partial z} \left(\mu \frac{\partial v_z}{\partial x} \right) \right\} - \frac{\partial}{\partial x} \left[\left(\frac{2}{3} \mu - \kappa \right) (\nabla \cdot \mathbf{v}) \right] \quad (195)$$

Similarly, for the remaining two directions:

$$\begin{aligned} \rho \frac{Dv_y}{Dt} = & -\frac{\partial p}{\partial y} + \nabla \cdot (\mu \nabla v_y) + \left\{ \frac{\partial}{\partial x} \left(\mu \frac{\partial v_x}{\partial y} \right) + \frac{\partial}{\partial y} \left(\mu \frac{\partial v_y}{\partial y} \right) + \frac{\partial}{\partial z} \left(\mu \frac{\partial v_z}{\partial y} \right) \right\} - \\ & - \frac{\partial}{\partial y} \left[\left(\frac{2}{3} \mu - \kappa \right) (\nabla \cdot \mathbf{v}) \right] + \rho F_y \end{aligned}$$

$$\begin{aligned} \rho \frac{Dv_z}{Dt} = & -\frac{\partial p}{\partial z} + \nabla \cdot (\mu \nabla v_z) + \left\{ \frac{\partial}{\partial x} \left(\mu \frac{\partial v_x}{\partial z} \right) + \frac{\partial}{\partial y} \left(\mu \frac{\partial v_y}{\partial z} \right) + \frac{\partial}{\partial z} \left(\mu \frac{\partial v_z}{\partial z} \right) \right\} - \\ & - \frac{\partial}{\partial z} \left[\left(\frac{2}{3} \mu - \kappa \right) (\nabla \cdot \mathbf{v}) \right] + \rho F_z \end{aligned}$$

In a more compact form:

$$\begin{aligned} \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) &= 0 \\ \rho \frac{D\mathbf{v}}{Dt} &= -\nabla p + \nabla \cdot (\mu \nabla \mathbf{v}) + \mathbf{S} + \rho \mathbf{F} \end{aligned}$$

Observation I Actually, the real Navier-Stokes' equations are the equations above written for a fluid with constant density and viscosity. Under these conditions, we have:

$$\begin{aligned} \nabla \cdot \mathbf{v} &= 0 \\ \rho \frac{D\mathbf{v}}{Dt} &= -\nabla p + \nabla \cdot (\mu \nabla \mathbf{v}) + \rho \mathbf{F} \end{aligned}$$

Indeed, because of the continuity equation $\mathbf{S}$ is equal to zero, as you can easily verify.

Observation II We can express explicitly the heat dissipated by the viscous forces that appears in the energy transport equation, directly using Newton's law. In particular, in Cartesian coordinates:

$$\begin{aligned} Q_{diss} = -\underline{\boldsymbol{\tau}} : \nabla \mathbf{v} = \mu \Phi_V = \mu \left\{ 2 \left[\left(\frac{\partial v_x}{\partial x} \right)^2 + \left(\frac{\partial v_y}{\partial y} \right)^2 + \left(\frac{\partial v_z}{\partial z} \right)^2 \right] + \left[\frac{\partial v_x}{\partial y} + \frac{\partial v_y}{\partial x} \right]^2 \right. \\ \left. + \left[\frac{\partial v_x}{\partial z} + \frac{\partial v_z}{\partial x} \right]^2 + \left[\frac{\partial v_y}{\partial z} + \frac{\partial v_z}{\partial y} \right]^2 - \left(\frac{2}{3} \mu - \kappa \right) \left[\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right]^2 \right\} \end{aligned}$$

8 Classification of flows

Let us go back to the equations of momentum. They are much more complex than they may seem: indeed they are strongly non-linear, coupled, and difficult to solve. numerically. Moreover, it is extremely difficult to prove mathematically that they admit a single solution with fixed boundary and initial conditions. Experience has shown that Navier-Stokes' equations are able to describe Newtonian fluids sufficiently accurately in the vast majority of cases. However, only in a limited number of cases (and in geometries of extreme simplicity) it is possible to arrive to their analytical solution. These cases are important for studying some of the fundamental features of fluid flows, but their relevance from the point of view of application is minimal.

In all cases where an analytical solution is not possible, it is necessary to proceed according to appropriate numerical methods. However in many cases of practical interest, some of the terms in the Navier-Stokes equations are null or negligible, because they are significantly less important than the rest. Obviously this simplification leads to a reduction in the mathematical complication of equations, but it also introduces errors. However, even these simplified equations can be solved analytically only in rare cases, but the numerical methods that need to be employed are often much simpler than those required for the complete set of equations. So let us have a look at the most important simplifications we can consider.

8.1 Incompressible flows

Incompressible flows can characterize both incompressible and compressible fluids (i.e. both liquid and gaseous fluids): in the first case the compressibility factor of a liquid is virtually null; in the second case, the incompressible flow hypothesis is only possible if fluid velocities are sufficiently lower than the propagation speed of the sound (in general it is required that the Mach number is less than 0.30). Special numerical algorithms can be applied when the hypothesis of incompressible flow is acceptable.

8.2 Inviscid flows (Euler's equations)

Generally at sufficiently high distances from the walls, the viscosity effects are very small. If we neglect the effects of viscosity, i.e. we assume the stress tensor is reduced to $\underline{\pi} = p\delta$, the equations of Navier-Stokes are reduced to so-called *Euler's equations*. The equation of continuity remains the same, but the equations of conservation of momentum become:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0 \quad (196)$$

$$\rho \frac{D\mathbf{v}}{Dt} = -\nabla p + \rho \mathbf{F} \quad (197)$$

Since the fluid is not viscous (we say it is inviscid), there is no friction on the walls, at which the fluid can flow without any resistance. Euler equations are used study high speed flows. In these conditions, due to high fluid velocity, viscous effects are important only in a region of very small thickness on solid surfaces and Euler equations are able to give a sufficiently accurate representation of the motion of the fluid. The Euler equations are not in themselves easier than the complete equations, since the nonlinear term (which is generally more problematic to deal with) remains the same. The advantage is that we can use much more coarse grids. Only the boundary layer requires a detailed description of the grid. For example, the study of wing profiles or aerodynamic aspects is performed through the Euler's equations.

8.3 Potential flows

Potential flows are the easiest flows to solve. They are simultaneously characterized by their non-viscous properties (as in Euler equations) and by the fact that the velocity field is irrotational:

$$\text{rot}(\mathbf{v}) = 0 \quad (198)$$

From this latter condition one can prove the existence of a velocity potential Φ , such that the velocity vector can be written as follows: $\mathbf{v} = -\nabla\Phi$.

At the same time, the equation of continuity, assuming a constant density, implies that $\nabla\mathbf{v} = 0$. Then combining the two equations, we get a single equation describing the potential Φ and hence allowing the velocity field to be determined:

$$\nabla \cdot (\nabla\mathbf{v}) = 0 \quad (199)$$

The above equation is a scalar equation (Laplace equation) and allows us to compute the potential over the computation domain. It can also be written by introducing the Laplace operator: $\nabla^2\mathbf{v} = 0$. Using the velocity definition $\mathbf{v} = -\nabla\Phi$ then the velocity field can be determined. It is true that Laplace's equation, in its simplicity, can not be resolved analytically unless in particular cases (and still with very simple geometries): however, its numerical resolution is extremely easy and does not have any particular difficulty. Next to the velocity potential, it is also possible to define a streamline function ψ , defined in such a way that the velocity vectors are tangent to the current lines (streamlines); in turn, the streamlines are orthogonal to the constant potential lines.

Potential flows are extremely interesting in research and theoretical, but they are not very realistic and therefore of little practical interest.

8.4 Viscous flows (Stokes' equations)

When the number of Reynolds that characterizes the motion of a fluid is very low (that is, when the fluid velocity is very low, the viscosity is high and/or the geometry size is very small) the convective (or nonlinear) term in the Navier-Stokes equations can be neglected. The flow is dominated by viscous, pressure, and external forces (*creeping flow*). If the fluid properties can be considered constant, the equations of conservation of momentum becomes linear and take the name of *Stokes equations*:

$$\frac{\partial\rho}{\partial t} + \nabla \cdot (\rho\mathbf{v}) = 0 \quad (200)$$

$$\frac{\partial(\rho\mathbf{v})}{\partial t} = -\nabla p - \nabla \cdot \underline{\underline{\tau}} + \rho\mathbf{F} \quad (201)$$

The main advantage is that the non linear term corresponding to convection was completely removed.

8.5 Boussinesq's approximation

In reactive flows or in general non-isothermal flows, fluid properties are a function of temperature, and in particular, the density can not be considered constant and uniform. However, if density variations are not very high, the density can be treated as if it were a constant in accumulation and convective terms, and its variability can only be maintained in the external term associated with the acceleration of gravity. This is known as

Boussinesq approximation, and consists in assuming that the density varies linearly with the temperature. Then the source term in the equation for conservation of momentum becomes:

$$(\rho - \rho_0) g_i = -\rho_0 g_i \beta (T - T_0) \quad (202)$$

where β is the coefficient of compressibility. This approximation does not involve any significant errors if the maximum temperature differences are in the order of $10 - 20^\circ C$. If the differences were to become higher, then errors could become significant.

8.6 Boundary-layer equations

If the motion of the fluid has a special direction of development (and therefore there are no recirculation) and the geometry does not show any sharp variation, the flow is almost exclusively influenced by the upstream fluid. Consider, for example, fluid motion in pipes, channels, flat or slightly curved surfaces. In these cases we may consider them as *boundary layer flows*. In such conditions, Navier Stokes's equations can be simplified, since:

- diffusive (or viscous) transport of momentum along the main direction is much smaller than convective transport and it can therefore be neglected;
- the velocity component along the main direction is considerably larger than the components in the other two directions;
- the pressure gradient component along the main direction is much greater than that in the other directions.

Without going into details, it can be shown that the boundary layer equations for a two-dimensional flow are reduced to the following form (in Cartesian coordinates):

$$\frac{\partial (\rho v_x)}{\partial t} + \frac{\partial (\rho v_x v_x)}{\partial x} + \frac{\partial (\rho v_y v_x)}{\partial y} = \mu \frac{\partial^2 v_x}{\partial y^2} - \frac{\partial p}{\partial x} \quad (203)$$

$$\frac{\partial p}{\partial y} = 0 \quad (204)$$

where x is the coordinate tangent to the solid surface and y the normal direction. The first equation corresponds to the conservation of momentum along the main direction, while the second equation the conservation of momentum along the normal direction. Of course, the equation of continuity must be added to this pair of equations. In this set of equations, however, there is no information about the pressure p as a function of the x coordinate. Well, this information must be provided by calculating the fluid flow outside the boundary layer (generally obtained through Euler equations or even under the potential flow hypothesis). The boundary layer equations can generally be solved by numerical methods much simpler than those needed to solve complete equations and are widely used in the field of aerodynamics.

8.7 Vorticity-Streamlines $\Omega - \psi$ Method

Two-dimensional incompressible flows with constant properties can be studied more easily by introducing the following functions: vorticity Ω and streamfunction ψ . For simplicity we refer to a system of orthogonal cartesian coordinates, for which the two functions are defined as follows:

$$\frac{\partial \psi}{\partial y} = v_x \quad \frac{\partial \psi}{\partial x} = -v_y \quad (205)$$

$$\Omega = \frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \quad (206)$$

The lines with constant ψ are directed parallel to the streamfunction lines. The vorticity Ω can actually be defined also for three-dimensional flows (this is a vector):

$$\mathbf{\Omega} = \nabla \times \mathbf{v} \quad (207)$$

In two-dimensional flows the vector $\mathbf{\Omega}$ is always orthogonal to the plane on which the motion is studied, and thus the above equation is reduced to (205). The main reason for introducing the Ω and ψ functions is that the equations describing the fluid flow can be written and solved in a much simpler way using Ω and ψ as independent variables, rather than velocity and pressure. If we actually replace the (205) equation in the vorticity definition (206), we get a kinematic equation that introduces a link between Ω and ψ :

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = -\Omega \quad (208)$$

If we now differentiate the equations of conservation of momentum in the x and y directions with respect to y and x and subtract the two equations resulting from one another, we get the transport equation of the vorticity:

$$\rho \frac{\partial \Omega}{\partial t} + \rho v_x \frac{\partial \Omega}{\partial x} + \rho v_y \frac{\partial \Omega}{\partial y} = \mu \left(\frac{\partial^2 \Omega}{\partial x^2} + \frac{\partial^2 \Omega}{\partial y^2} \right) \quad (209)$$

It is clear that the pressure does not appear in any of the two equations (208) and (209), that is, it has been deleted as a dependent variable. Therefore, the equations of Navier-Stokes have been replaced by a pair of partial equations (208) and (209), instead of the three classical equations (two for the components v_x and v_y and one for pressure). It is the reduction of the number of variables and especially the absence of pressure between the dependent variables which makes the application of this method particularly advantageous. If necessary, pressure can be obtained by solving a suitable Poisson equation (this aspect will be better discussed later). Obviously, the equations (208) and (209), predict the Ω and ψ variables. From these it is easy to get the components v_x and v_y of velocity through the equation (208).

The disadvantage of this method is related to the difficulties in defining the boundary conditions, especially in complex geometries. For example, solid contours and symmetry planes are surfaces characterized by a constant value of ψ (since the flow is parallel to those surfaces). Unfortunately, however, the ψ values are not known on such surfaces and can only be determined during the calculation, by knowledge of v_x and v_y . The problem is even more significant for the Ω vorticity, because neither its value nor its derivatives are known *a priori* on the boundaries, but they can only be determined during the calculation from ψ . This leads to a significant reduction in the convergence speed of numerical methods. Particularly critical becomes the definition of boundary conditions in the corners of a computational domain; it is necessary in such cases to consider highly sophisticated and complex approaches.

Despite the reported difficulties, the Ω - ψ method is still useful for studying two-dimensional flows. In principle, it could also be extended to three-dimensional flows, but from the numerical point of view this is never done.

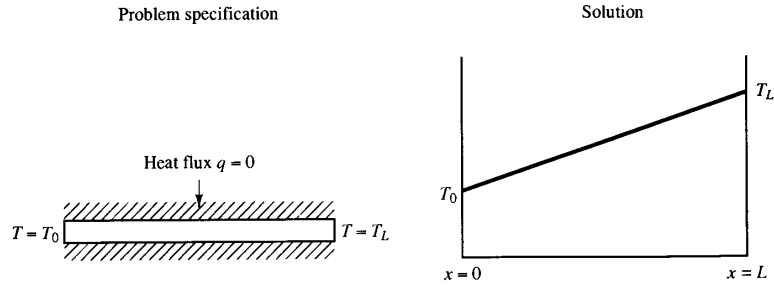


Figure 13: Steady-state heat diffusion in a bar.

9 Classification of Transport Equations

The conservation and transport equations derived in the previous sections require initial conditions (IC) and boundary conditions (BC) to be solved. Defining these conditions is a delicate process and must be done with great care. It's not always easy or even possible to express initial and boundary conditions accurately, and often significant simplifications are required. The manner in which these conditions are defined largely depends on the characteristics of the equations to which they apply. In particular, two categories of problems can be distinguished: stationary problems (steady-state or equilibrium problems) and non-stationary problems (unsteady or marching problems, although the latter term is more general).

9.1 Steady-state (stationary) problems

As implied by the name, problems of this type do not involve the time dependence of the considered variables and thus represent physical situations where a temporal transient has been surpassed, leading to a state of equilibrium. Classical stationary problems in fluid mechanics are governed by elliptic equations. The simplest elliptic equation, which in some ways can be considered a prototype for more general elliptic equations, is the Laplace equation (which, for instance, describes the potential of an irrotational Eulerian flow):

$$\nabla^2 \Phi = 0 \quad (210)$$

where Φ represents a generic variable. The Laplace equation can describe numerous physical phenomena, such as thermal conduction in solids. In Figure 13, purely for illustrative purposes, we have the case of a metal bar with assigned temperatures at the ends, constant over time. The problem is one-dimensional and is governed by a Laplace equation: $\frac{\partial^2 T}{\partial x^2} = 0$. This equation can only be solved if boundary conditions for the dependent variable (in this case, T) are specified on all boundaries of the computational domain. For this reason, problems of this type, requiring boundary conditions on all boundaries, are also referred to as boundary-value problems.

A highly significant characteristic of elliptic problems is that a disturbance at any point within the computational domain or on its boundaries leads to a change in the solution at all other points. Therefore, disturbances are said to propagate in all spatial directions. Additionally, the solution of a problem governed by an elliptic equation never exhibits discontinuities; it remains smooth even when discontinuities might be present in the boundary conditions. This aspect is of great importance for the formulation and application of numerical methods to solve the problem. To ensure that information can propagate in all

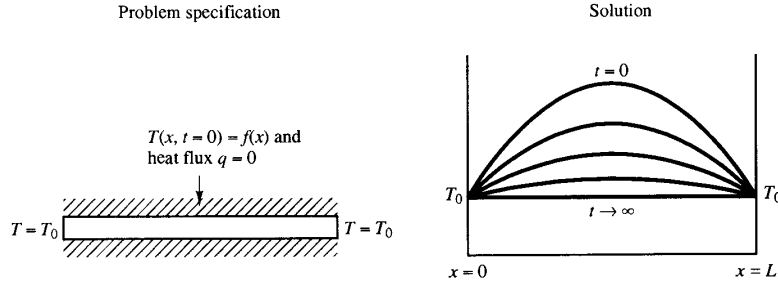


Figure 14: Thermal diffusion problem in a non-steady-state metal bar.

spatial directions, numerical techniques for elliptic problems must establish a connection between variables at a given point on the computational grid and the variables defined at nearby points.

9.2 Unsteady problems

Non-steady problems naturally involve a temporal evolution of various quantities characterizing fluid motion. Non-steady problems in fluid mechanics can be either parabolic or hyperbolic. Strictly speaking, a marching problem doesn't necessarily imply a non-steady problem. Indeed, some steady motions can be described by parabolic or hyperbolic equations. Consider, for example, the boundary layer equations, which are derived under the assumption of the existence of a privileged coordinate for fluid motion that behaves equivalently to a temporal coordinate, even though it is spatial. For simplicity, we won't delve into such problems, and thus, we will use the term "non-steady" (though not entirely accurately) when referring to parabolic or hyperbolic equations.

Parabolic equations describe non-steady problems that involve significant energy dissipation. The prototype equation of this kind is the diffusion equation:

$$\frac{\partial \Phi}{\partial t} = \alpha \nabla^2 \Phi \quad (211)$$

Let's consider a simple example, once again in one dimension. Let's revisit the metal bar from the previous example and assume that both ends are held at a constant temperature T_0 . Furthermore, let's assume that at the initial time $t = 0$, the temperature profile is parabolic, as shown in Figure 14. Starting from this moment, if the bar is isolated, it begins to cool down until it reaches the steady-state solution, which is a uniform temperature of T_0 . This problem is governed by an equation similar to the one we wrote earlier, which in one-dimensional terms is: $\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$, where α is the thermal diffusivity. It's clear that for this type of problem, we not only need to specify boundary conditions but also an initial condition. Problems of this kind are therefore called initial-boundary-value problems.

A disturbance at any point in the computational domain or on the boundary can influence the solution throughout the computational domain, but only for times greater than the time at which the disturbance occurs. The solution is said to move forward in time and spread in space. Thanks to the presence of dissipative phenomena, the solution is always continuous within the computational domain, even if the initial and boundary conditions are discontinuous. The steady-state solution is rigorously reached as $t \rightarrow \infty$, and it is an elliptic solution. In fact, if we set the non-steady term $\frac{\partial \Phi}{\partial t}$ to zero in (211), we obtain an elliptic equation.

Hyperbolic-type equations characterize non-stationary problems in which dissipative effects are negligible. The prototype of hyperbolic equations is the wave equation:

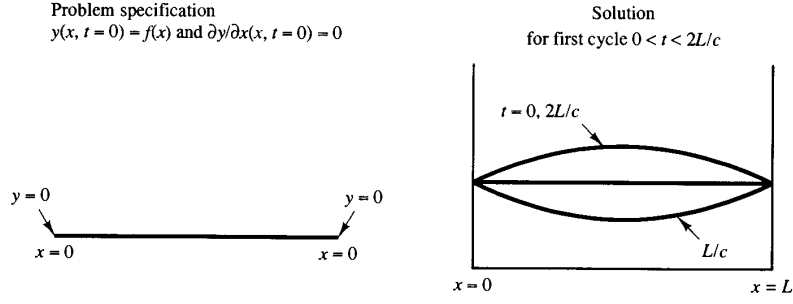


Figure 15: Vibrations of a tensioned string.

$$\frac{\partial^2 \Phi}{\partial t^2} = c^2 \nabla^2 \Phi \quad (212)$$

An equation of this type, for example, describes the vibrations of a string with fixed endpoints in space. The constant c is called the wave velocity. The solution of equation (212) in one dimension, with a representing the initial amplitude, is as follows (Figure 15):

$$\Phi(x, t) = a \cos\left(\frac{\pi ct}{L}\right) \sin\left(\frac{\pi x}{L}\right) \quad (213)$$

This solution shows that the amplitude of the vibration remains constant due to the absence of dissipative terms. The absence of such phenomena has important consequences. The equations that govern the motion of fluids at high velocities (with $\text{Mach} \sim 1$ or higher) have a hyperbolic character, manifested through the so-called shock waves (which ultimately are discontinuities). Numerical techniques for solving such problems must be able to describe possible discontinuities in the solution within the computational domain. In particular, the presence of a disturbance at a certain point in the computational domain can influence only certain regions of it, unlike what happens with elliptic and parabolic equations. In fact, the disturbance propagation velocity is equal to c and is therefore finite, unlike elliptic and parabolic equations, which assume an infinite disturbance velocity.

9.3 Classification of Equations

In general, equations describing fluid behavior can have mixed characteristics, being parabolic in some regions of space and hyperbolic in others. There is a straightforward method to classify partial differential equations (PDEs) when working in 2D. Consider the general equation:

$$a \frac{\partial^2 \Phi}{\partial x^2} + b \frac{\partial^2 \Phi}{\partial x \partial y} + c \frac{\partial^2 \Phi}{\partial y^2} + d \frac{\partial \Phi}{\partial x} + e \frac{\partial \Phi}{\partial y} + f \Phi + g = 0 \quad (214)$$

Assume, for simplicity, that the coefficients a, b, c, d, e, f, g are constants, and that the equation is linear. The classification of the PDE depends only on the highest-order derivatives, so let's focus on the second-order derivatives. It can be shown that equation (214) admits hyperbolic solutions only if its characteristic equation has two real roots:

$$a \left(\frac{dy}{dx}\right)^2 - b \left(\frac{dy}{dx}\right) + c = 0 \quad (215)$$

Ultimately, this property depends on the value of the discriminant $(b^2 - 4ac)$. There are three possible cases, summarized in Table 4.

$(b^2 - 4ac)$	Equation Type	Characteristics
> 0	Hyperbolic	Two real roots
$= 0$	Parabolic	One real root
< 0	Elliptic	No real roots

Table 4: Classification of second-order linear PDEs.

This simple method can be extended to cases where the coefficients a, b, c, d, e, f, g are not constants and when the equation (214) is not linear. However, in these situations, as mentioned earlier, the character of the PDE can vary in different regions of space. For example, consider the following equation:

$$y \frac{\partial^2 \Phi}{\partial x^2} + \frac{\partial^2 \Phi}{\partial x^2} = 0 \quad (216)$$

Let's focus on the region $-1 < y < 1$, where $a = a(x, y) = y$, $b = 0$, $c = 1$. The value of the discriminant $(b^2 - 4ac)$ is $-4y$. We must distinguish three cases:

- If $y < 0$: $(b^2 - 4ac) > 0$, and thus, the equation is hyperbolic.
- If $y = 0$: $(b^2 - 4ac) = 0$, and thus, the equation is parabolic.
- If $y > 0$: $(b^2 - 4ac) < 0$, and thus, the equation is elliptic.

So, equation (216) is mixed; it can be locally parabolic, hyperbolic, or elliptic depending on the value of y . The classification of such equations depends on the local values of the coefficients a, b, c .

This discussion can also be extended to partial differential equations in N independent variables $(x_1, x_2, \dots, x_N)$. The starting point is writing these equations in the general form:

$$\sum_j^N \sum_k^N A_{jk} \frac{\partial^2 \Phi}{\partial x_j \partial x_k} + H = 0 \quad (217)$$

where $A_{jk} = A_{kj}$. It can be shown that it is sufficient to examine the eigenvalues λ of the matrix A_{jk} to understand the characteristics of the equation. Thus, it is sufficient to calculate the eigenvalues λ by solving the characteristic polynomial equation:

$$\det [A_{jk} - \lambda I_{jk}] = 0 \quad (218)$$

The classification rules are straightforward:

- If at least one eigenvalue $\lambda = 0$, the equation is parabolic.
- If all eigenvalues $\lambda \neq 0$ and have the same sign, the equation is elliptic.
- If all eigenvalues $\lambda \neq 0$, and at least one of them has a different sign from the others, the equation is hyperbolic.

Obviously, the criterion formulated for equations in two independent variables is a particular case of this more general criterion.

In the case of *systems* of partial differential equations, things become more complicated, but the underlying ideas for classification are very similar to what we have just discussed. At least in the case of transport equations for reactive systems, the common practice is to formally classify the equations as shown in Table 5, despite the fact that the actual behavior of fluids is much more complex.

Flow Type	Steady Conditions	Non-Steady Conditions
Navier-Stokes Equations	Elliptic	Parabolic
Euler Equations ($M < 1$)	Elliptic	Hyperbolic
Euler Equations ($M > 1$)	Hyperbolic	Hyperbolic
Boundary Layer Equations	Parabolic	Parabolic

Table 5: Classification of major flow types.

Initial Conditions	
Throughout the computational domain, the values of ρ , $\mathbf{v}$, T , ω_j at time $t = 0$ must be specified.	
Boundary Conditions	
Solid Surfaces	$\mathbf{v} = \mathbf{v}_{wall}$ (no-slip condition)
	Values of T and ω_j or their fluxes must be specified.
Inlets and Outlets	Inlet: ρ , $\mathbf{v}$, T , ω_j must be specified as functions of position.
	Outlet: $-p + \mu \frac{\partial v_n}{\partial n} = F_n$ and $\mu \frac{\partial v_t}{\partial n} = F_t$ (stress continuity)

Table 6: Initial and boundary conditions for transport equations.

10 Definition of Boundary Conditions

Based on what was emphasized in the previous section, it is clear that defining boundary conditions is not a straightforward task due to the complicated and mixed (parabolic, hyperbolic, and elliptic) nature of transport equations. In fact, there are no general theoretical (purely mathematical) guidelines to determine admissible boundary conditions. Common practice is to identify the most appropriate boundary conditions based on physical considerations. Table 6 simplifies the common initial and boundary conditions used for the Navier-Stokes equations and the transport equations of species and energy. The subscripts n and t indicate the directions normal and tangential to the boundary, and F represents the assigned stress on the surface.

Note that it is not necessary to specify the density value on the surfaces or outlets, thanks to the particular nature of the continuity equation. On the contrary, at the inlets, this value is required. In all other points within the computational domain and on all other boundaries, the density is the result of the continuity equation and does not require any boundary condition to be specified (of course, assuming the incompressibility of the fluid, the density is constant and uniform, so the problem does not arise entirely).

The most commonly used conditions for outflows are to assign the values of the normal and tangential stress:

$$-p + \mu \frac{\partial v_n}{\partial n} = F_n \quad (219)$$

$$\mu \frac{\partial v_t}{\partial n} = F_t \quad (220)$$

Outflows are generally located where the fluid motion can be approximated as unidirectional, and where it is possible to know the stresses on the boundary in advance. For example, for high Reynolds numbers at large distances from solid boundaries (in external flows) or in the fully developed flow region (in internal flows), it is known

in advance that velocities in the direction perpendicular to the boundary no longer change. Thus, under these assumptions, one has $F_n = -p$ and $F_t = 0$.

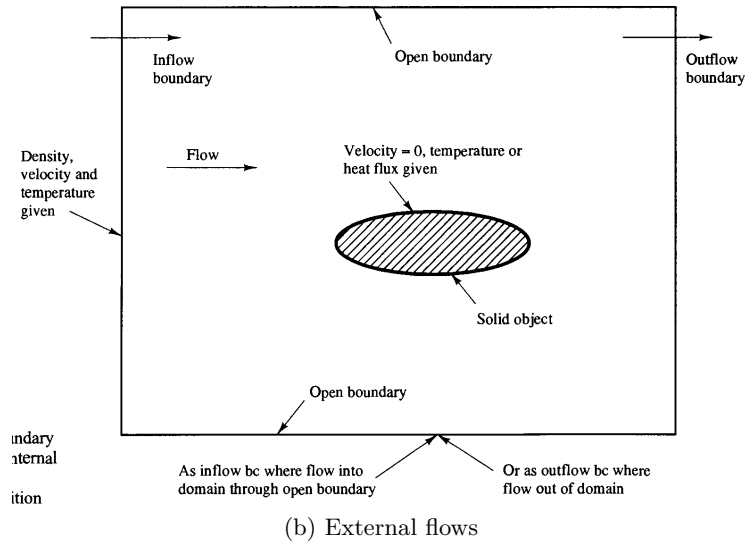
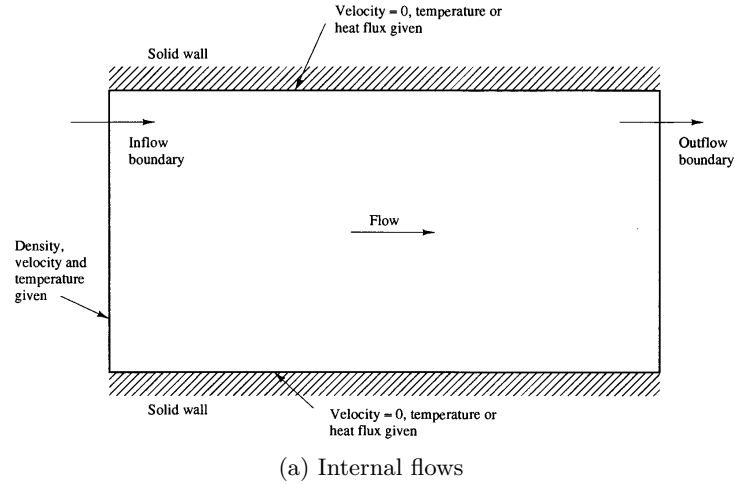


Figure 16: Typical boundary conditions for transport equations

There are certainly more accurate and conceptually correct conditions than these, but practice has shown that equation (219) is able to balance simplicity of implementation with a good description of reality. Therefore, except for special cases, it is commonly considered. Figure 16 shows examples of typical boundary conditions for internal and external flows.

Often it is possible to impose symmetry and periodicity conditions to take advantage of specific geometric characteristics of the computational domain (Figure 17):

- symmetry: $\frac{\partial \Phi}{\partial n} = 0$;
- periodicity: $\Phi_1 = \Phi_2$.

Suggested readings

- **J.H. Ferziger, M. Peric, R.L. Street**, *Computational Methods for Fluid Dynamics*, Springer (2020)

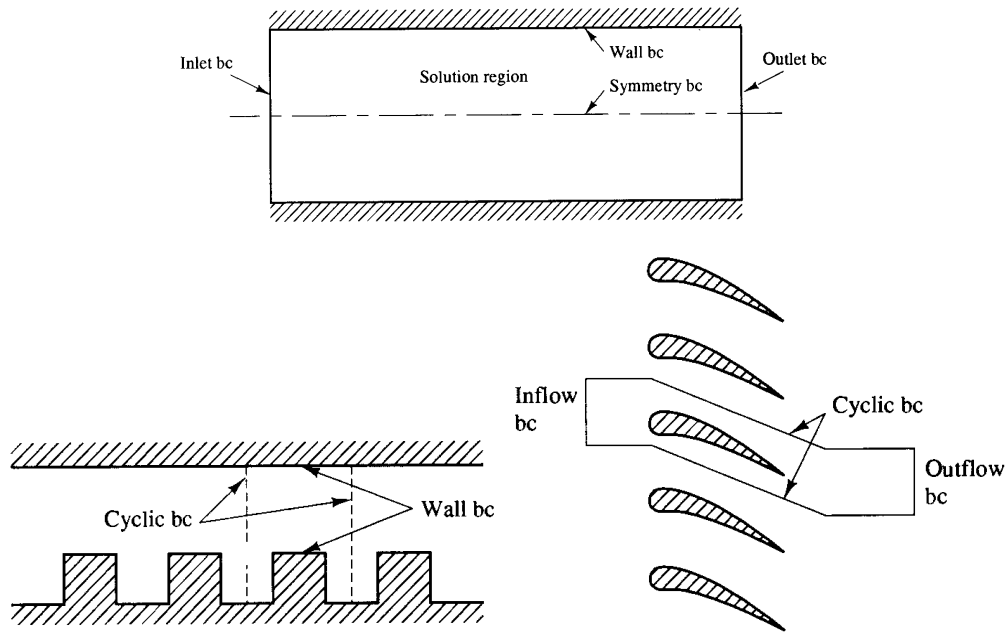


Figure 17: Examples of symmetric and periodic boundaries.

- **H. Versteeg, M. Malalasekera**, *Introduction to Computational Fluid Dynamics - The Finite Volume Method*, Pearson (2007)
- **C.J. Greenshields, H. Weller**, *Introduction Notes on Computational Fluid Dynamics: General Principles*, CFD Direct Limited (2022)